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GWSpace: a multi-mission science data simulator for space-based gravitational wave detection

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Abstract

Space-based gravitational wave detectors such as TianQin, laser interferometer space antenna (LISA), and Taiji have the potential to outperform themselves through joint observation. To achieve this, it is desirable to practice joint data analysis in advance on simulated data that encodes the intrinsic correlation among the signals found in different detectors that operate simultaneously. In this paper, we introduce **GWSpace**, a package that can simulate the joint detection data from TianQin, LISA, and Taiji. The software is not a groundbreaking work that starts from scratch. Rather, we use as many open-source resources

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as possible, tailoring them to the needs of simulating the multi-mission science data and putting everything into a ready-to-go and easy-to-use package. We shall describe the main components, the construction, and a few examples of application of the package. A common coordinate system, namely the Solar System barycenter coordinate system, is utilized to calculate spacecraft orbits for all three missions. The paper also provides a brief derivation of the detection process and outlines the general waveform of sources detectable by these detectors.

Keywords: gravitational wave, detector response, space-based gravitational wave detection

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1. Introduction

Several space-based gravitational wave (GW) detectors, including TianQin [1], the laser interferometer space antenna (LISA) [2, 3], and Taiji [4, 5], are eyeing for launch around mid-2030s. These detectors will, for the first time, open the unexplored milli-Hertz (mHz) frequency band of GW spectrum. In complement to the current ground-based GW detectors (GBDs) [6], space-based GW detectors (SBDs) enjoy a plethora of new types of sources, including the galaxy compact binary (GCB) [7, 8], the massive black hole binary (MBHB) [9], the stellar-mass black hole binary (SBHB) [10, 11], the Extreme Mass Ratio Inspirals (EMRI) [12, 13], the stochastic GW background (SGWB) [14, 15], and etc [16–21].

Unlike GBDs that mainly capture GW events in their short-lived merger phases, SBDs detect GW events mostly during their long-lasting inspiral phases, resulting in complex data sets with overlapping signals in time and frequency. Consequently, this poses significant challenges to data analysis [11, 22–24]. So several mock data challenges, such as mock LISA data challenge (MLDC) [22, 25, 26], which is now replaced by LISA data challenge (LDC) [23], and Taiji data challenge (TDC) [24], have been set up to help develop the necessary tools need for space-based GW data analysis.

It is possible that more than one of the detectors, TianQin, LISA, and Taiji, will be observing concurrently during the mid-2030s, enabling a network approach to detect some GW signals. These detectors can then observe the same GW signals from different locations in the Solar System, effectively forming a virtual detector with a much larger size [27], leading to significant improvements in sky localization accuracy [28–32], allowing for the discovery of more sources and a deeper understanding of physics [33, 34]. More examples showing how joint detection can improve over individual detectors can be found in [7, 9, 10, 12, 29, 32, 35]. A comprehensive study of how the joint detection with TianQin and LISA can improve over each detector can be found in [36]. What's more, the difficulties faced by space-based GW data analysis are partially due to parameter degeneracy [37], and it has been shown that joint detection can also be helpful here by breaking some of the degeneracies [38]. So it is important to seriously consider the possibility of doing joint data analysis from different SBD combinations.

There are challenges in conducting data analysis for joint observations with multiple detectors. For example, due to the significant differences in arm lengths and orbits, approximations and optimized algorithms developed for geocentric and heliocentric cannot be directly applied interchangeably. What's more, variations in the separations among the detectors will affect the correlation of the signals, and this requires comprehensive consideration in the calculation of the likelihood and covariance matrices. To facilitate the study of problems involved in the analysis of joint observational data, we introduce in this paper [GWSpace](#), which is a package that can simulate the joint simulation data from all three SBDs mentioned above.

Although MLDC, LDC, and TDC have already achieved simulating data for individual detectors like LISA and Taiji, there are new problems to be solved when one wants to simulate data for all three detectors operating together. For example, due to the shorter arm-length of TianQin, its sensitivity frequency band is more shifted toward the higher frequency end, ranging from 10^{-4} to 1 Hz [1], as compared to about $[2 \times 10^{-5}, 10^{-1}]$ Hz for LISA and Taiji [3, 5]. Because of this, the response model derived using the low-frequency limit method [39] that works for LISA and Taiji is not always valid for TianQin. Therefore, it is necessary to consider the full-frequency response models to accurately describe the behavior of all the detectors across the entire frequency spectrum [40, 41]. Another issue is that one needs to study the response of the three SBDs by using the same coordinate system to correctly reveal the correlation among them. The Solar System barycenter (SSB) coordinate system is identified as the most straightforward choice for this purpose.

The paper is structured as follows. Section 2 specifies the coordinate systems used in this paper. Section 3 specifies the orbits of the three SBDs involved, namely TianQin, LISA and Taiji. Sections 4–6 detail the response, TDI combinations, and source waveforms used in [GWSpace](#). Some example data sets are described in section 7. A summary is in section 8.

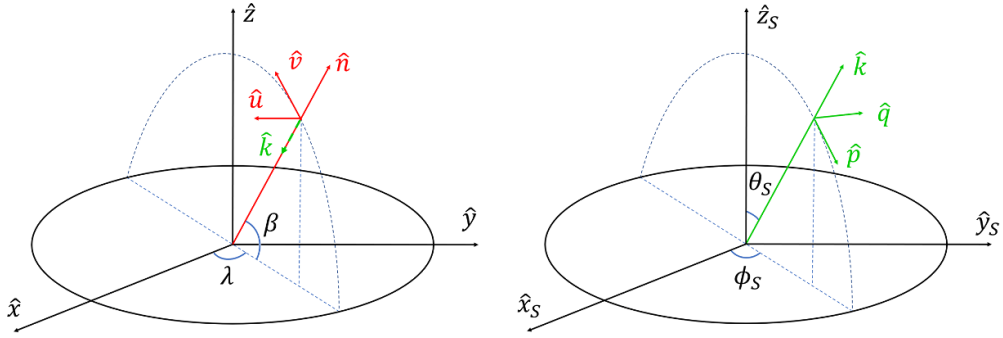


Figure 1. (Left) The detector frame and the relative orientation between $(\hat{x}, \hat{y}, \hat{z})$ and $(\hat{n}, \hat{u}, \hat{v})$. (Right) The source frame and the relative orientation between $(\hat{x}_S, \hat{y}_S, \hat{z}_S)$ and $(\hat{k}, \hat{p}, \hat{q})$.

2. Coordinate systems

Two basic coordinate systems will be used in this paper: the astronomical ecliptic coordinate system used to describe the detector, hence called the detector frame, and the coordinate system adapted to the description of GW sources, hence called the source frame.

The detector frame, as illustrated in figure 1(Left), is defined with the origin at the SSB. In this frame, the z -axis is oriented perpendicular to the ecliptic and points towards the north, while the x -axis points towards the March equinox. The y -axis is obtained as $\mathbf{y} = \mathbf{z} \times \mathbf{x}$. The direction to a GW source is indicated with the unit vector $\hat{n} = \hat{n}(\lambda, \beta)$, where λ and β are the celestial longitudes and celestial latitude of the source, respectively. To describe the polarization of GWs propagating along $\hat{k} = -\hat{n}$, two additional auxiliary unit vectors are introduced⁸

$$\hat{u} = \frac{\hat{n} \times \hat{z}}{|\hat{n} \times \hat{z}|} = \frac{\hat{z} \times \hat{k}}{|\hat{z} \times \hat{k}|}, \quad \hat{v} = \hat{u} \times \hat{n} = \hat{k} \times \hat{u}, \quad (1)$$

so that the trio, $(\hat{u}, \hat{v}, \hat{k})$, forms a right-handed orthogonal basis. Then, one can obtain that

$$\hat{u} = [\sin \lambda, -\cos \lambda, 0], \quad (2)$$

$$\hat{v} = [-\sin \beta \cos \lambda, -\sin \beta \sin \lambda, \cos \beta], \quad (3)$$

$$\hat{k} = -\hat{n} = [-\cos \beta \cos \lambda, -\sin \lambda \cos \beta, -\sin \beta]. \quad (4)$$

The source frame is illustrated in figure 1(Right). Exactly how the origin and the axes, $(\hat{x}_S, \hat{y}_S, \hat{z}_S)$, are chosen for the source dynamics will be determined on a case-by-case basis. The direction to the GW detector is indicated with the unit vector $\hat{k} = \hat{k}(\theta_S, \phi_S)$. To describe the polarization of GWs propagating along $\hat{k}$, two auxiliary unit vectors are also introduced,

$$\hat{q} = \frac{\hat{z} \times \hat{k}}{|\hat{z} \times \hat{k}|}, \quad \hat{p} = \hat{q} \times \hat{k}, \quad (5)$$

⁸ <https://lisa-ldc.lal.in2p3.fr/static/data/pdf/LDC-manual-002.pdf>.

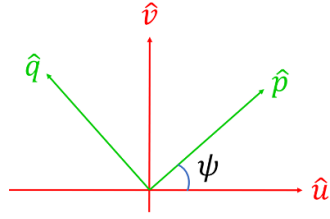


Figure 2. Relative orientations of $(\hat{u}, \hat{v})$ and $(\hat{p}, \hat{q})$. Note that $\hat{k}$ is perpendicular to and coming out of the paper.

so that the trio, $(\hat{p}, \hat{q}, \hat{k})$, forms a right-handed orthogonal basis. Then, one can obtain that

$$\hat{p} = [\cos \iota \cos \varphi, \sin \varphi \cos \iota, -\sin \iota], \quad (6)$$

$$\hat{q} = [-\sin \varphi, \cos \varphi, 0], \quad (7)$$

$$\hat{k} = [\sin \iota \cos \varphi, \sin \iota \sin \varphi, \cos \iota]. \quad (8)$$

Since $\hat{n} = -\hat{k}$, the planes spanned by $(\hat{u}, \hat{v})$ and $(\hat{p}, \hat{q})$ are parallel to each other (see figure 2). As a result,

$$\begin{aligned} \hat{p} &= \cos \psi \hat{u} + \sin \psi \hat{v}, & \hat{q} &= -\sin \psi \hat{u} + \cos \psi \hat{v}, \\ \hat{u} &= \cos \psi \hat{p} - \sin \psi \hat{q}, & \hat{v} &= \sin \psi \hat{p} + \cos \psi \hat{q}. \end{aligned} \quad (9)$$

Thus, the polarization angle can be computed as

$$\psi = \arctan_2 [\hat{p} \cdot \hat{u}, \hat{p} \cdot \hat{v}]. \quad (10)$$

Using the basis vectors, one can define the polarization tensors in the source frame and the SSB frame. In the source frame

$$e_{ij}^+ = (\hat{p} \otimes \hat{p} - \hat{q} \otimes \hat{q})_{ij}, \quad e_{ij}^\times = (\hat{p} \otimes \hat{q} + \hat{q} \otimes \hat{p})_{ij}. \quad (11)$$

Similarly, in the SSB frame

$$\epsilon_{ij}^+ = (\hat{u} \otimes \hat{u} - \hat{v} \otimes \hat{v})_{ij}, \quad \epsilon_{ij}^\times = (\hat{u} \otimes \hat{v} + \hat{v} \otimes \hat{u})_{ij}. \quad (12)$$

After some calculation, the relation between the polarization tensors can be rewritten as

$$e^+ = \epsilon^+ \cos 2\psi + \epsilon^\times \sin 2\psi, \quad (13)$$

$$e^\times = -\epsilon^+ \sin 2\psi + \epsilon^\times \cos 2\psi. \quad (14)$$

In the source frame, the GW strain in a transverse-traceless gauge takes the form

$$h_{ij}^{TT} = e_{ij}^+ h_+ + e_{ij}^\times h_\times, \quad (15)$$

where $h_{+, \times}$ are the plus and cross mode of GW. The corresponding representation of the strain in the SSB frame is

$$\begin{aligned} h_{ij}^{TT} &= h_+ (\epsilon^+ \cos 2\psi + \epsilon^\times \sin 2\psi)_{ij} + h_\times (\epsilon^\times \cos 2\psi - \epsilon^+ \sin 2\psi)_{ij} \\ &= (h_+ \cos 2\psi - h_\times \sin 2\psi) \epsilon_{ij}^+ + (h_+ \sin 2\psi + h_\times \cos 2\psi) \epsilon_{ij}^\times. \end{aligned} \quad (16)$$

Then, in the SSB frame, one can define two new modes of GW as

$$h_+^{\text{SSB}} = h_+ \cos 2\psi - h_\times \sin 2\psi, \quad (17)$$

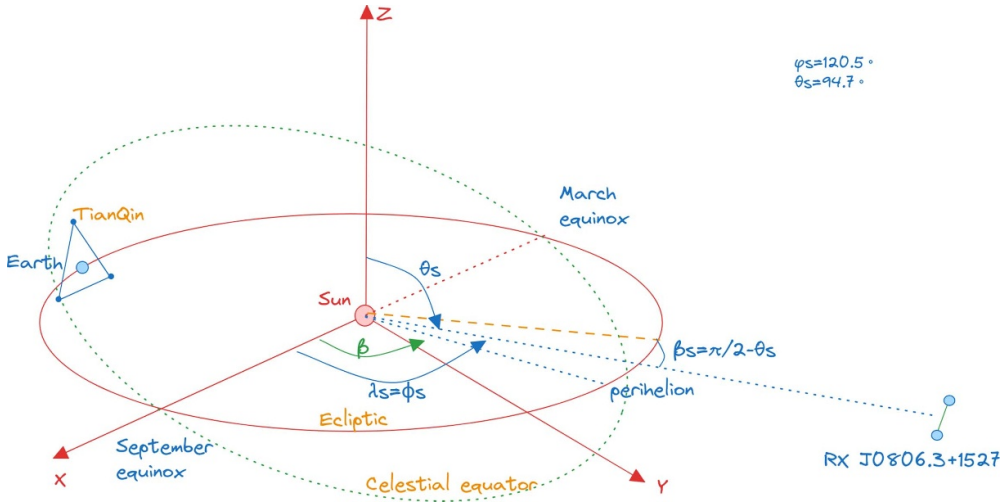


Figure 4. Schematic of the TianQin spacecraft's orbit for TianQin in the SSB coordinate system.

$$Z(t) = 0, \quad (21)$$

where $\alpha = 2\pi f_m t + \kappa_0$, $f_m = 1/(\text{one sidereal year}) = 3.14 \times 10^{-8} \text{ Hz}$ is the orbit modulation frequency, κ_0 is the mean ecliptic longitude measured from the vernal equinox (or September equinox) at $t = 0$, and β denotes the angle measured from the vernal equinox to the perihelion.

In the context of the TianQin spacecraft's motion around the Earth, its orbits remain consistent with equation (21). However, when considering the SSB frame, TianQin assumes a specific orientation towards the direction of J0806 ($\{\lambda_s, \beta_s\} = \{120.5^\circ, -4.7^\circ\}$), as shown in figure 5). Introducing a coordinate system rotation and disregarding eccentricity, the description of TianQin's orbits can be further refined [43]

$$x_n = \frac{L}{\sqrt{3}} [\sin \beta_s \cos \lambda_s \sin(\alpha_n - \beta') + \sin \lambda_s \cos(\alpha_n - \beta')], \quad (22)$$

$$y_n = \frac{L}{\sqrt{3}} [\sin \beta_s \sin \lambda_s \sin(\alpha_n - \beta') - \cos \lambda_s \cos(\alpha_n - \beta')], \quad (23)$$

$$z_n = -\frac{L}{\sqrt{3}} \cos \beta_s \sin(\alpha_n - \beta'), \quad (24)$$

where $L = \sqrt{3} R_{\text{tq}}$ is the arm-length between the two spacecraft, $\alpha(t) = 2\pi f_{\text{sc}} t + \kappa_n$, $\kappa_n = +\frac{2}{3}\pi(n-1) + \lambda$, λ is the initial orbit phase of the first ($n=1$) spacecraft measured from $\tilde{x}$ axis, $f_{\text{sc}} = 1/\sqrt{GM_{\text{Earth}}/R_{\text{tq}}^3} \simeq 1/(3.65 \text{ day})$ is the modulation frequency due to the rotation of the detector around the guiding center, β' is the angle measured from the $\tilde{x}$ axis to the perigee of the first spacecraft orbit. Here, we assume the three spacecraft are in circular orbits around the Earth, thus, β' will be some arbitrary number (one can just set it as 0).

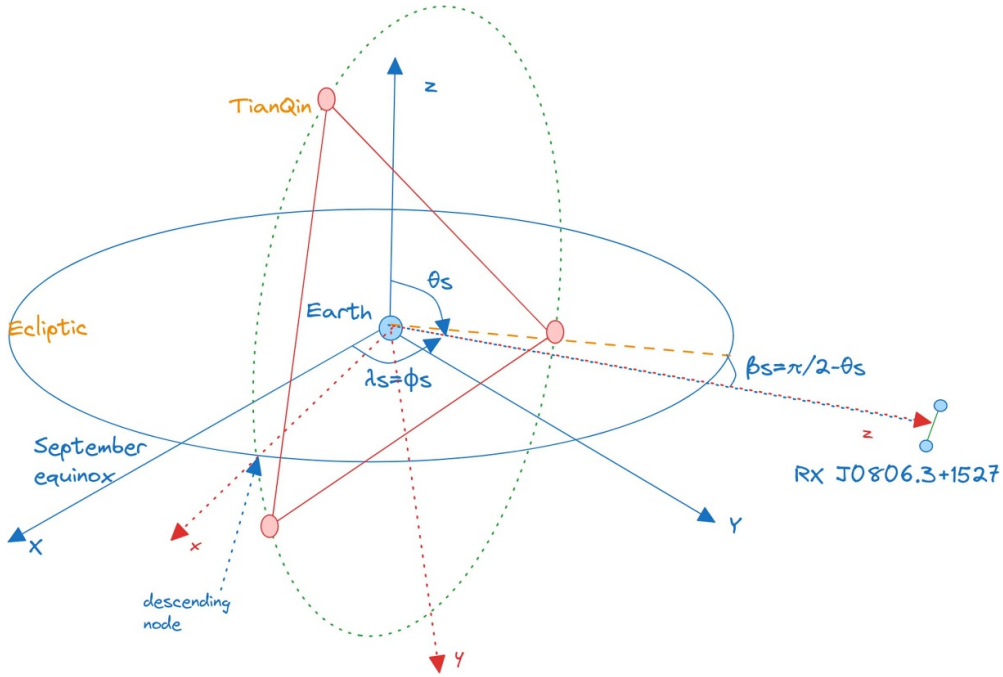


Figure 5. Schematic of the detector coordinate system $\{\tilde{x}, \tilde{y}, \tilde{z}\}$ and the geocentric-ecliptic coordinate system $\{x, y, z\}$. $\tilde{x}$ point to the descending node, $\tilde{z}$ axis points to J0806.

3.2. LISA and Taiji: heliocentric orbit

According to the description in [44], when considering a constellation of spacecraft in individual Keplerian orbits with an inclination of $i = \sqrt{e}$, the coordinates of each spacecraft can be elegantly expressed in the following form (this expression has been expanded up to the second order of eccentricity) [44]

$$x_n = a \cos(\alpha'') + ae (\sin \alpha'' \cos \alpha'' \sin \beta'_n - (1 + \sin^2 \alpha'') \cos \beta'_n) + \frac{1}{8} ae^2 (3 \cos(3\alpha'' - 2\beta'_n) - 10 \cos \beta'_n - 5 \cos(\alpha'' - 2\beta'_n)), \quad (25)$$

$$y_n = a \sin(\alpha'') + ae (\sin \alpha'' \cos \alpha'' \cos \beta'_n - (1 + \cos^2 \alpha'') \sin \beta'_n) + \frac{1}{8} ae^2 (3 \sin(3\alpha'' - 2\beta'_n) - 10 \sin \alpha'' + 5 \sin(\alpha'' - 2\beta'_n)), \quad (26)$$

$$z_n = -\sqrt{3}ae \cos(\alpha'' - \beta'_n) + \sqrt{3}ae^2 [1 + \sin^2(\alpha'' - \beta'_n)]. \quad (27)$$

Here $a = R_{\text{LISA,TJ}} = 1$ AU is the radial distance to the guiding center for LISA and Taiji, $\alpha'' = \alpha - \beta \mp 20^\circ$ for LISA and Taiji, where α and β are the same as those in Earth orbit or equations (19)–(21). And $\beta'_n = \frac{2\pi}{3}(n-1) + \lambda'$, λ' is the initial orientation of the constellation, $e \simeq L_{\text{LISA,TJ}}/(2a\sqrt{3})$ represent the orbital eccentricity, $L_{\text{LISA}} = 2.5 \times 10^6$ km and $L_{\text{TJ}} = 3 \times 10^6$ km is the arm-length between two spacecraft for LISA and Taiji, respectively.

While the spacecraft orbits for LISA and Taiji are situated in the ecliptic planes, their constellations' guiding center follows nearly circular trajectories. In the GWSpace code, the perihelion angle of the three spacecraft for LISA and Taiji is set to be the same as that of Earth.

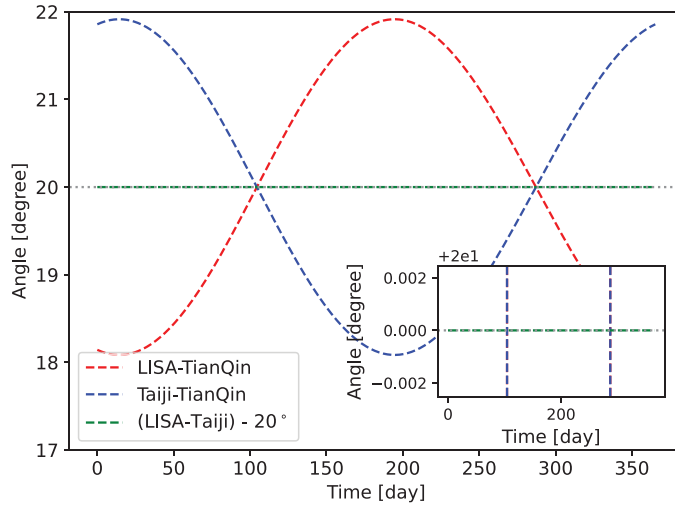


Figure 6. The relative angle between different detectors. Here, for improved visual clarity, the angle between LISA and Taiji has been adjusted by subtracting 20 degrees.

However, TianQin's guiding center coincides with Earth's, resulting in the changing angle between LISA and Taiji over time. Figure 6 illustrates the relative angles between the different detectors. It can be observed that the angle between LISA or Taiji and Earth varies between 18° and 22° , while the angle between LISA and Taiji is approximately 40° , with a slight variation of around 2.4×10^{-3} . These findings are consistent with the proposed orbit described in reference [3, 5].

4. Detector response

In a vacuum, propagating GWs induces a time-varying strain in the fabric of space-time. This strain can alter the proper distance between freely falling masses, providing a means to gather information about the GWs. One approach is to measure the variation in light travel time or optical path length between two test masses [45]. As a GW passes through, these separated masses will experience relative acceleration or tilting. Consequently, a GW detector is employed to monitor the separation between the test masses. There are two commonly used methods to monitor the distance between two objects: radar ranging or similar techniques, and measuring the Doppler shift in a signal transmitted from one object to the other [45]. However, a question arises regarding whether the GW affects the electromagnetic waves used for measuring distances [45]. In the following sections, we will provide a brief overview of how a GW detector responds to GW signals.

4.1. The general waveform and mode decomposition

Assuming a Universe consisting solely of vacuum and GW. Since GWs are very weak, the metric of the spacetime perturbed by a GW can be described as

$$ds^2 = -c^2 dt^2 + [\delta_{ij} + h_{ij}(t)] dx^i dx^j, \quad (28)$$

where h_{ij} is the tensor perturbation, it is directly related to the GW itself, carrying information about its amplitude, frequency, and polarization. By analyzing the changes in the metric caused

by the GW, we can extract valuable information about the GW signal. In the TT coordinate system (with coordinates $x^0 = ct, x^1 = x, x^2 = y, x^3 = z$), a weak GW can be described as a weak plane wave traveling in the $+z$ direction. The line element describing the metric of spacetime in this scenario is given by

$$ds^2 = -c^2 dt^2 + \left(1 + h_+ \left(t - \frac{z}{c}\right)\right) dx^2 + \left(1 - h_+ \left(t - \frac{z}{c}\right)\right) dy^2 + 2h_\times \left(t - \frac{z}{c}\right) dx dy + dz^2. \quad (29)$$

General waveform The GW can be approximated as an arbitrary plane wave with wave vector $\vec{k}$ and a Tensor form ‘amplitude’, thus

$$\mathbf{h}(t, \mathbf{r}) = \mathbf{h}_0 e^{i(2\pi f t - \vec{k} \cdot \mathbf{r}/c)} = \mathbf{h}_0 e^{i2\pi f(t - \hat{k} \cdot \mathbf{r}/c)} = h(t - \hat{k} \cdot \mathbf{r}) = h(\xi), \quad (30)$$

where $\hat{k} = \frac{\vec{k}}{|\vec{k}|} = \frac{\vec{k}}{2\pi f}$ is the propagation direction of GW, $\mathbf{r}$ is an arbitrary direction, $\xi = t - \hat{k} \cdot \mathbf{r}$ is a surface of constant phase.

There is relative motion between the source frame and the detector frame. In the detector frame, the SSB is moving relative to the cosmic microwave background (CMB) with a peculiar velocity $v \approx 370 \text{ km s}^{-1} \approx 0.0012$, along the direction $\lambda \approx 172^\circ$, $\beta \approx -11^\circ$ [46]. In the source frame, the velocities of the sources can be introduced as model parameters.

Mode decomposition In the source frame, the GW can be further decomposed using spin-weighted spherical harmonics [47] $_{-2}Y_{\ell m}(\iota, \varphi)$ as

$$h_+ - ih_\times = \sum_{\ell \geq 2} \sum_{m=-\ell}^{\ell} {}_{-2}Y_{\ell m}(\iota, \varphi) h_{\ell m}. \quad (31)$$

where $\{\iota, \varphi\}$ represent the inclination and phase describing the orientation of emission. The primary harmonic is h_{22} , while the others are called higher harmonics or higher modes. And each mode can be described as

$$h_{\ell m} = A_{\ell m} e^{-i\Phi_{\ell m}}. \quad (32)$$

Based on this decomposition, we obtain

$$h_+ = \frac{1}{2} \sum_{\ell, m} (-2Y_{\ell m}(\iota, \varphi) h_{\ell m} + {}_{-2}Y_{\ell m}^*(\iota, \varphi) h_{\ell m}^*), \quad (33)$$

$$h_\times = \frac{i}{2} \sum_{\ell, m} (-2Y_{\ell m}(\iota, \varphi) h_{\ell m} - {}_{-2}Y_{\ell m}^*(\iota, \varphi) h_{\ell m}^*). \quad (34)$$

In particular, for the non-processing binary systems with a fixed equatorial plane of orbit, there exists an exact symmetry relation between modes

$$h_{\ell, -m} = (-1)^m h_{\ell m}^*. \quad (35)$$

With this symmetry, one has

$$h_{+, \times} = \sum_{\ell, m} K_{\ell m}^{+, \times} h_{\ell m}, \quad (36)$$

where

$$K_{\ell m}^+ = \frac{1}{2} \left(-2Y_{\ell m} + (-1)^m {}_{-2}Y_{\ell, -m}^* \right), \quad (37)$$

$$K_{\ell m}^{\times} = \frac{i}{2} \left(-{}_2Y_{\ell m} - (-1)^{\ell} {}_2Y_{\ell, -m}^{*} \right). \quad (38)$$

It is convenient to introduce mode-by-mode polarization matrices

$$P_{\ell m} = e_{+} K_{\ell m}^{+} + e_{\times} K_{\ell m}^{\times}, \quad (39)$$

so that the GW signal in matrix form will be

$$\mathbf{h}^{TT} = \sum_{\ell, m} P_{\ell m} h_{\ell m}. \quad (40)$$

In the SSB frame, one can write

$$P_{+} + iP_{\times} = e^{-i2\psi} (\epsilon_{+} + i\epsilon_{\times}). \quad (41)$$

With the above equations, $P_{\ell m}$ will be

$$\begin{aligned} P_{\ell m}(\iota, \varphi, \psi) &= \frac{1}{2} {}_2Y_{\ell m}(\iota, \varphi) e^{-i2\psi} (\epsilon_{+} + i\epsilon_{\times}) \\ &\quad + \frac{1}{2} (-1)^{\ell} {}_2Y_{\ell, -m}^{*}(\iota, \varphi) e^{+i2\psi} (\epsilon_{+} - i\epsilon_{\times}). \end{aligned} \quad (42)$$

In this way, we can factor out explicitly all dependencies in the extrinsic parameters (ι, φ, ψ) .

Suppose that the GW only has the main mode, i.e. the 22 mode, we have $h_{22} = A_{22} e^{-i\Phi_{22}}$ and $h_{2, -2} = h_{22}^{*} = A_{22} e^{i\Phi}$. The expressions of the spin-weighted spherical harmonics for the mode of $\{2, \pm 2\}$ are

$${}_2Y_{22}(\iota, \varphi) = \frac{1}{2} \sqrt{\frac{5}{\pi}} \cos^4 \frac{\iota}{2} e^{i2\varphi}, \quad {}_2Y_{2, -2}(\iota, \varphi) = \frac{1}{2} \sqrt{\frac{5}{\pi}} \sin^4 \frac{\iota}{2} e^{-i2\varphi}, \quad (43)$$

and

$$\begin{aligned} K_{22}^{+} &= \frac{1}{2} ({}_2Y_{22}(\iota, \varphi) + {}_2Y_{2, -2}^{*}(\iota, \varphi)) \\ &= \frac{1}{4} \sqrt{\frac{5}{\pi}} \left(\cos^4 \frac{\iota}{2} + \sin^4 \frac{\iota}{2} \right) e^{i2\varphi} \\ &= \frac{1}{4} \sqrt{\frac{5}{\pi}} \frac{(1 + \cos^2 \iota)}{2} e^{i2\varphi}, \end{aligned} \quad (44)$$

$$\begin{aligned} K_{22}^{\times} &= \frac{i}{2} ({}_2Y_{22}(\iota, \varphi) - {}_2Y_{2, -2}^{*}(\iota, \varphi)) \\ &= \frac{i}{4} \sqrt{\frac{5}{\pi}} \left(\cos^4 \frac{\iota}{2} - \sin^4 \frac{\iota}{2} \right) e^{i2\varphi} \\ &= -\frac{i}{4} \sqrt{\frac{5}{\pi}} \cos \iota e^{i2\varphi}. \end{aligned} \quad (45)$$

and so

$$K_{2, -2}^{+} = (K_{22}^{+})^{*}, \quad K_{2, -2}^{\times} = (K_{22}^{\times})^{*}. \quad (46)$$

Thus, one has

$$h_{+} = K_{22}^{+} h_{22} + K_{2, -2}^{+} h_{2, -2} = A_{22} \sqrt{\frac{5}{4\pi}} \frac{1 + \cos^2 \iota}{2} \cos(\Phi_{22} - 2\varphi), \quad (47)$$

$$h_{\times} = K_{22}^{\times} h_{22} + K_{2, -2}^{\times} h_{2, -2} = A_{22} \sqrt{\frac{5}{4\pi}} \cos \iota \sin(\Phi_{22} - 2\varphi). \quad (48)$$

For non-precessing systems, equation (35) will be translate to

$$\tilde{h}_{\ell,-m}(f) = (-1)^\ell \tilde{h}_{\ell m}(-f)^* \quad (49)$$

in the Fourier domain. For a given mode of GW waveform, one has $h_{\ell m} \propto \exp[-im\phi_{\text{orbit}}]$, where ϕ_{orbit} is the orbital phase of the GW systems, and it always verifies with $\dot{\phi}_{\text{orbit}} > 0$. Thus, for non-precessing systems or in the processing frame for a binary with misaligned spins, an approximation often applied as

$$\tilde{h}_{\ell m}(f) \simeq 0 \quad \text{for } m > 0, \quad f > 0, \quad (50)$$

$$\tilde{h}_{\ell m}(f) \simeq 0 \quad \text{for } m < 0, \quad f < 0, \quad (51)$$

$$\tilde{h}_{\ell 0}(f) \simeq 0. \quad (52)$$

In this way, for the positive frequencies $f > 0$, $\tilde{h}_{+, \times} = \sum_{\ell} \sum_{m < 0} K_{\ell, m}^{+, \times} \tilde{h}_{\ell m}$.

Eccentric mode decomposition Eccentric waveforms also generate the harmonics, which act similarly to higher modes but are described by the mean orbital frequency. Under the stationary phase approximation (SPA), there is a relationship between the mean orbital frequency F and the Fourier frequency f for different eccentric harmonics:

$$f = j \cdot F(t_0). \quad (53)$$

Here we use the index j to distinguish eccentric harmonics from spin-weighted spherical harmonics above. t_0 is the time that gives the stationary point of F . The dominant eccentric harmonic is $j=2$.

With $(\ell, m) = (2, 2)$, a frequency domain eccentric waveform can be written as

$$\tilde{h}_{+, \times} = \sum_{j=1}^{10} \tilde{\mathcal{A}}_j \xi_j^{+, \times} e^{-i\Psi_j}. \quad (54)$$

Here

$$\xi_j^{+, \times} = C_{+, \times}^{(j)} + iS_{+, \times}^{(j)}, \quad (55)$$

which is a function of (ι, φ) and the eccentricity $e(F)$. When $e = 0$,

$$\xi_{j=2}^+ = C_+^{(2)} + iS_+^{(2)} = 4 \cdot \frac{1 + \cos^2 \iota}{2} e^{i \cdot 2\varphi}, \quad (56)$$

$$\xi_{j=2}^\times = C_\times^{(2)} + iS_\times^{(2)} = 4 \cdot (-\cos \iota) e^{i \cdot 2\varphi}, \quad (57)$$

$$\xi_{j \neq 2}^{+, \times} = 0, \quad (58)$$

which go back to the coefficients of the dominant mode $(\ell, m) = (2, 2)$ [48]. But for a non-zero eccentricity, one cannot explicitly write $P_{\ell m}$ as we shown in equation (42), and should directly use $P_+, P_\times$ in equation (41).

4.2. Single arm response in time domain

The effect of GWs on matter can be described as a tidal deformation. To detect the GW, one method is to test the distance changes between two spatially separated free-falling test masses.

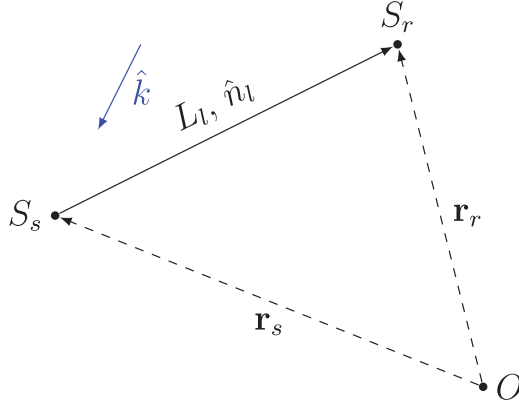


Figure 7. A radio signal send from the point S_s travels along the arm L_l in the direction of $\hat{n}_l$ towards the receiver at S_r . The coordinate origin is denoted by O , while point S_s and point S_r are located at $\mathbf{r}_s$ and $\mathbf{r}_r$, respectively.

Suppose that the photon travels along the direction of test mass 1 (S_s) to test mass 2 (S_r) as $\hat{n}_l$, as shown in figure 7. It follows a null geodesic, i.e. $ds^2 = 0$. Thus, the metric reads

$$cdt = \sqrt{(\delta_{ij} + h_{ij}(\xi)) dx^i dx^j}, \quad (59)$$

where

$$\xi(l) = t(l) - \hat{k} \cdot \mathbf{r}(l)/c = t_s + l/c - \hat{k} \cdot [\mathbf{r}_s(t_s) + \hat{n}(t_s) l]/c, \quad (60)$$

$\mathbf{r}_s$ is the position of S_s , $\mathbf{r}(l)$ is the position of photon at time t , $l = \sqrt{\sum_i (x^i - x_s^i)^2} = |\mathbf{r}(l) - \mathbf{r}_s|$, $\hat{n} = \frac{\mathbf{r}(l) - \mathbf{r}_s}{l}$, and

$$\begin{aligned} \frac{d\xi}{dl/c} &= 1 - \hat{k} \cdot \hat{n}_l(t_s), \quad \frac{dx^i}{dl} = \hat{n}^i, \quad \hat{n}^i \hat{n}_i = 1 \\ &\Rightarrow \\ \frac{dx^i/c}{d\xi} &= \frac{dx^i}{dl} \frac{dl/c}{d\xi} = \frac{\hat{n}^i}{1 - \hat{k} \cdot \hat{n}} \end{aligned} \quad (61)$$

With the above derivation, equation (59) can be rewritten as

$$\begin{aligned} dt &= \sqrt{(\delta_{ij} + h_{ij}(\xi)) \frac{dx^i/c}{d\xi} \frac{dx^j/c}{d\xi}} d\xi = \sqrt{1 + h_{ij} \hat{n}^i \hat{n}^j} \frac{d\xi}{1 - \hat{k} \cdot \hat{n}} \\ &\approx \left(1 + \frac{1}{2} h_{ij} \hat{n}^i \hat{n}^j + \mathcal{O}(h^2) \right) \frac{d\xi}{1 - \hat{k} \cdot \hat{n}}. \end{aligned} \quad (62)$$

Then, from S_s to S_r , the duration of the proper time will be

$$\begin{aligned} \int_{t_s}^{t_r} dt &= \int_0^{L_l} \sqrt{1 + h_{ij} \hat{n}^i \hat{n}^j} \frac{d\xi}{1 - \hat{k} \cdot \hat{n}} \approx \int_0^{L_l} \left(1 + \frac{1}{2} \hat{n}_l^T \cdot \mathbf{h} \cdot \hat{n}_l + \mathcal{O}(h^2) \right) \frac{d\xi}{1 - \hat{k} \cdot \hat{n}_l} \\ &= \int_0^{L_l} dl/c + \int_0^{L_l} \frac{\hat{n}_l^T \cdot \mathbf{h} \cdot \hat{n}_l}{2(1 - \hat{k} \cdot \hat{n}_l)} d\xi, \end{aligned} \quad (63)$$

where L_l is the length between S_s and S_r , $\hat{n}_l$ is the unit vector of the photon propagation.

Here, if one supposes that the position of S_s and S_r does not change or changes very little during the photon moving from S_s to S_r , which means $\hat{n}_l(t_s) \approx \hat{n}_l(t_r) = \hat{n}_l$. Then, for simplicity, the integral in equation (63) can be rewritten as

$$t_r = t_s + L_l/c + \frac{1}{2(1 - \hat{k} \cdot \hat{n}_l)} \hat{n}_l^T \cdot \left(\int_{\xi_s}^{\xi_r} \mathbf{h}(\xi) d\xi \right) \cdot \hat{n}_l. \quad (64)$$

From this equation, one can directly get the path length fluctuations due to the GW

$$\delta l_{sr}(t) = \frac{c}{2(1 - \hat{k} \cdot \hat{n}_l)} \hat{n}_l^T \cdot \left(\int_{\xi_s}^{\xi_r} \mathbf{h}(\xi) d\xi \right) \cdot \hat{n}_l. \quad (65)$$

Suppose the frequency of the photon is not changed during the photon's travel from S_s to S_r . Then the total phase change of the photon will be $\phi_{\text{tot}} = 2\pi\nu_0(t_r - t_s)$. If there is no GW, the phase change will be $\phi_{\text{ori}} = 2\pi\nu_0 L/c$. So, with the help of equation (64), the phase fluctuations measured under the GW will be

$$\Delta\phi(t) = \phi_{\text{tot}} - \phi_{\text{ori}} = 2\pi\nu_0 \delta l_{sr}(t)/c. \quad (66)$$

To get the time of reception changes concerning the time of emission, one can differentiate the above equation for t_s

$$\begin{aligned} \frac{dt_r}{dt_s} &= 1 + \frac{1}{2(1 - \hat{k} \cdot \hat{n}_l)} \hat{n}_l^T \cdot \left(\int_{\xi_0}^{\xi_l} \frac{d\mathbf{h}(\xi)}{d\xi} \frac{d\xi}{dt_s} d\xi \right) \cdot \hat{n}_l \\ &= 1 + \frac{1}{2(1 - \hat{k} \cdot \hat{n}_l)} \hat{n}_l^T \cdot [\mathbf{h}(\xi_r) - \mathbf{h}(\xi_s)] \cdot \hat{n}_l. \end{aligned} \quad (67)$$

Here, we have used the assumption that the motion of S_s and S_r is much slower compared to the time of laser beam propagation, i.e. $d\mathbf{r}_s/dt_s \approx 0$ and $d\hat{n}_l/dt_s \approx 0$, so $d\xi/dt_s = 1$.

The interferometers used to detect GWs do not emit single photons but continuous lasers with frequency $\nu(t)$. If the phase change of the photon at S_s and S_r are the same, we have $d\phi/(2\pi) = \nu_s dt_s = \nu_r dt_r$. Then one can get the dimensionless fractional frequency deviation $y^{\text{GW}}(t)$ as

$$y_{sr}^{\text{GW}}(t_r) = \frac{\nu_r - \nu_s}{\nu_s} = \frac{\nu_r}{\nu_s} - 1 = \frac{d\phi/dt_r}{d\phi/dt_s} - 1 \quad (68)$$

$$= \frac{1}{1 + \frac{1}{2(1 - \hat{k} \cdot \hat{n}_l)} \hat{n}_l^T \cdot [\mathbf{h}(\xi_r) - \mathbf{h}(\xi_s)] \cdot \hat{n}_l} - 1 \quad (69)$$

$$\approx \frac{1}{2(1 - \hat{k} \cdot \hat{n}_l)} \hat{n}_l^T \cdot [\mathbf{h}(\xi_s) - \mathbf{h}(\xi_r)] \cdot \hat{n}_l + \mathcal{O}(h^2). \quad (70)$$

Hence

$$\begin{aligned} \xi_s &= t_s - \hat{k} \cdot \mathbf{r}_s(t_s)/c \approx t_r - L_l/c - \hat{k} \cdot [\mathbf{r}_s(t_r - L_l/c)]/c \\ &\approx t_r - L_l/c - \hat{k} \cdot [\mathbf{r}_s(t_r) - \partial_t \mathbf{r}_s(t_r) L_l/c]/c \\ &\approx t_r - L_l/c - \hat{k} \cdot \mathbf{r}_s(t_r)/c, \end{aligned} \quad (71)$$

$$\xi_r = t_r - \hat{k} \cdot \mathbf{r}_r(t_r)/c. \quad (72)$$

In the third line, we assumed that $\partial_t \mathbf{r}_s \ll c$. Finally, one has the relative frequency deviation at the time of $t = t_r$ as [49–51]

$$y_{\text{slr}}^{\text{GW}}(t) = \frac{1}{2(1 - \hat{k} \cdot \hat{n}_l)} \hat{n}_l^T \cdot \left[\mathbf{h}(t - L_l/c - \hat{k} \cdot \mathbf{r}_s/c) - \mathbf{h}(t - \hat{k} \cdot \mathbf{r}_r/c) \right] \cdot \hat{n}_l. \quad (73)$$

When the photon reflected from S_r to S_s , we have

$$y_{\text{rls}}^{\text{GW}}(t) = \frac{1}{2(1 + \hat{k} \cdot \hat{n}_l)} \hat{n}_l^T \cdot \left[\mathbf{h}(t - L_l/c - \hat{k} \cdot \mathbf{r}_r/c) - \mathbf{h}(t - \hat{k} \cdot \mathbf{r}_s/c) \right] \cdot \hat{n}_l. \quad (74)$$

Considering that the GW is described in the SSB coordinate, thus, one can redefine some parameters as

$$y_{\text{slr}}^{\text{GW}}(t) = \frac{1}{2(1 - \hat{k} \cdot \hat{n}_l)} \left[H(t - L/c - \hat{k} \cdot \mathbf{r}_s/c) - H(t - \hat{k} \cdot \mathbf{r}_r/c) \right], \quad (75)$$

where

$$\begin{aligned} H(t) &= n_l^i h_{ij}(t) n_l^j = n_l^i \left(h_+ \epsilon_{ij}^+ + h_\times \epsilon_{ij}^\times \right) n_l^j \\ &= n_l^i \left[h_+ (u_i u_j - v_i v_j) + h_\times (u_i v_j + v_i u_j) \right] n_l^j \\ &= h_+ \left(n_l^i u_i u_j n_l^j - n_l^i v_i v_j n_l^j \right) + h_\times \left(n_l^i u_i v_j n_l^j + n_l^i v_i u_j n_l^j \right) \\ &= h_+ \left[(\hat{n}_l \cdot \hat{u})^2 - (\hat{n}_l \cdot \hat{v})^2 \right] + h_\times \cdot 2 (\hat{n}_l \cdot \hat{u}) (\hat{n}_l \cdot \hat{v}) \\ &= h_+ \zeta_l^+ + h_\times \zeta_l^\times, \end{aligned} \quad (76)$$

and

$$\zeta_l^+ = \hat{n}_l \cdot \epsilon^+ \cdot \hat{n}_l = (\hat{n}_l \cdot \hat{u})^2 - (\hat{n}_l \cdot \hat{v})^2, \quad (77)$$

$$\zeta_l^\times = \hat{n}_l \cdot \epsilon^\times \cdot \hat{n}_l = 2 (\hat{n}_l \cdot \hat{u}) (\hat{n}_l \cdot \hat{v}). \quad (78)$$

For the two-way response, one can get [52]

$$\begin{aligned} y_{\text{sls}}^{\text{GW}}(t) &= \frac{\nu}{\nu_0} - 1 = \frac{\nu}{\nu'} \frac{\nu'}{\nu_0} - 1 = [y_{\text{slr}}^{\text{GW}}(t - L_l/c) + 1] [y_{\text{rls}}^{\text{GW}}(t) + 1] - 1 \\ &\approx y_{\text{slr}}^{\text{GW}}(t - L/c) + y_{\text{rls}}^{\text{GW}}(t) + \mathcal{O}(h^2) \\ &= \frac{1}{2} \left\{ (1 + \hat{k} \cdot \hat{n}) [\Psi_l(t - 2L_l/c) - \Psi_l(t - L/c)] \right. \\ &\quad \left. + (1 - \hat{k} \cdot \hat{n}) [\Psi_l(t - L_l/c) - \Psi_l(t)] \right\} \\ &= \frac{1 + \hat{k} \cdot \hat{n}}{2} \Psi_l(t - 2L_l/c) - \hat{k} \cdot \hat{n} \Psi_l(t - L_l/c) - \frac{1 - \hat{k} \cdot \hat{n}}{2} \Psi_l(t), \end{aligned} \quad (79)$$

where (using $\hat{n} = \hat{n}(t)$ for convenient)

$$\Psi_l(t') = \frac{\hat{n}_l^T \cdot \mathbf{h}(t' - \hat{k} \cdot \mathbf{r}(t')/c) \cdot \hat{n}_l}{1 - (\hat{k} \cdot \hat{n}_l)^2}. \quad (80)$$

However, one should note that the above derivation is based on the assumption that the positions of the spacecraft change very little between the photon sent from S_s to S_r .

4.3. Single arm response in frequency domain

Adopting the Fourier transform, the GW in the frequency domain will be [53]

$$h_0(t, \vec{x}) = h(t - d(t)) = \int df e^{i2\pi f(t-d(t))} \tilde{h}(f) \text{ or } h(\xi) = \int df e^{i2\pi f\xi} \tilde{h}(f). \quad (81)$$

With the Fourier transform, the path length fluctuations could be rewritten as

$$\begin{aligned} \delta l_{\text{sr}}(t) &= \frac{c}{2(1 - \hat{k} \cdot \hat{n}_l)} \hat{n}_l^T \cdot \left(\int_{\xi_s}^{\xi_r} d\xi \int df e^{i2\pi f\xi} \tilde{\mathbf{h}}(f) \right) \cdot \hat{n}_l \\ &= \frac{c}{2(1 - \hat{k} \cdot \hat{n}_l)} \hat{n}_l^T \cdot \left(\int df (e^{i2\pi f\xi_r} - e^{i2\pi f\xi_s}) \frac{\tilde{\mathbf{h}}(f)}{i2\pi f} \right) \cdot \hat{n}_l \\ &= L_l \hat{n}_l^T \cdot \left(\int df e^{i2\pi ft} \mathcal{T}_{\text{sr}}(\hat{k}, f, t) \tilde{\mathbf{h}}(f) \right) \cdot \hat{n}_l \end{aligned} \quad (82)$$

where $\mathcal{T}_{\text{sr}}(\hat{k}, f, t)$ is the transfer function [53]

$$\begin{aligned} \mathcal{T}_{\text{sr}}(f, t) &= \frac{c/L}{2(1 - \hat{k} \cdot \hat{n}_l)} \frac{1}{i2\pi f} \left(e^{-i2\pi f \hat{k} \cdot \mathbf{r}_r/c} - e^{-i2\pi f (L_l + \hat{k} \cdot \mathbf{r}_s)/c} \right) \\ &= \frac{c/L}{2(1 - \hat{k} \cdot \hat{n}_l)} \frac{1}{i2\pi f} \left(e^{i\pi f [L - \hat{k} \cdot (\mathbf{r}_r - \mathbf{r}_s)]/c} - e^{-i\pi f [L_l - \hat{k} \cdot (\mathbf{r}_r - \mathbf{r}_s)]/c} \right) \\ &\quad \times e^{-i\pi f [L + \hat{k} \cdot (\mathbf{r}_r + \mathbf{r}_s)]/c} \\ &= \frac{c/L}{2(1 - \hat{k} \cdot \hat{n})} \frac{i2 \sin\left(\pi f L/c (1 - \hat{k} \cdot \hat{n})\right)}{i2\pi f} e^{-i\pi f [L + \hat{k} \cdot (\mathbf{r}_r + \mathbf{r}_s)]/c} \\ &= \frac{1}{2} \text{sinc}\left(\pi f L/c (1 - \hat{k} \cdot \hat{n})\right) \exp\left\{-i\pi f [L + \hat{k} \cdot (\mathbf{r}_r + \mathbf{r}_s)]/c\right\}. \end{aligned} \quad (83)$$

Finally, one can define the one-arm detector tensor as

$$\mathbf{D}(\hat{k}, f, t) = \hat{n}(t) \otimes \hat{n}(t) \mathcal{T}(\hat{k}, f, t), \quad (84)$$

and the path length fluctuation in the frequency domain will be

$$\frac{\delta \tilde{l}_{\text{sr}}(f)}{L_l} = \mathbf{D}(\hat{k}, f, t) : \tilde{\mathbf{h}}(f), \quad (85)$$

where $(\hat{n} \otimes \hat{n})_{ij} = \hat{n}_i \hat{n}_j$, $\mathbf{A} : \mathbf{B} = A_i B_i$. Similarly, the phase fluctuation in the frequency domain will be

$$\Delta \tilde{\phi}(f) = \frac{2\pi \nu_0}{c} \delta \tilde{l}_{\text{sr}}(f). \quad (86)$$

Similarly, one can derive the relative frequency derivation in the frequency domain with the Fourier transform, which reads

$$\begin{aligned}
y_{\text{slr}}^{\text{GW}}(t) &= \frac{1}{2(1 - \hat{k} \cdot \hat{n}_l)} \hat{n}_l^T \cdot \left[\int df' e^{i2\pi f' (t - L_l/c - \hat{k} \cdot \mathbf{r}_s/c)} \mathbf{h}(f') \right. \\
&\quad \left. - \int df'' e^{i2\pi f'' (t - \hat{k} \cdot \mathbf{r}_r/c)} \mathbf{h}(f'') \right] \cdot \hat{n}_l \\
&= \frac{1}{2(1 - \hat{k} \cdot \hat{n}_l)} \hat{n}_l^T \cdot \int df e^{i2\pi f t} \tilde{\mathbf{h}}(f) \left[e^{-i2\pi f (L_l + \hat{k} \cdot \mathbf{r}_s)/c} - e^{-i2\pi f \hat{k} \cdot \mathbf{r}_r/c} \right] \cdot \hat{n}_l \\
&= \int df e^{i2\pi f t} \left[e^{-i2\pi f (L_l + \hat{k} \cdot \mathbf{r}_s)/c} - e^{-i2\pi f \hat{k} \cdot \mathbf{r}_r/c} \right] \frac{\hat{n}_l^T \cdot \tilde{\mathbf{h}}(f) \cdot \hat{n}_l}{2(1 - \hat{k} \cdot \hat{n}_l)} \\
&= \int df e^{i2\pi f t} \left[e^{-i\pi f [L_l + \hat{k} \cdot (\mathbf{r}_s - \mathbf{r}_r)]/c} - e^{-i\pi f (\hat{k} \cdot \mathbf{r}_r - L_l - \hat{k} \cdot \mathbf{r}_s)/c} \right] \\
&\quad \times e^{-i\pi f [L_l + \hat{k} \cdot (\mathbf{r}_s + \mathbf{r}_r)]/c} \frac{\hat{n}_l^T \cdot \tilde{\mathbf{h}}(f) \cdot \hat{n}_l}{2(1 - \hat{k} \cdot \hat{n}_l)} \\
&= \int df e^{i2\pi f t} \left\{ -\frac{i \sin \left[\pi f L_l/c (1 - \hat{k} \cdot \hat{n}_l) \right]}{(1 - \hat{k} \cdot \hat{n}_l)} e^{-i\pi f [L_l + \hat{k} \cdot (\mathbf{r}_s + \mathbf{r}_r)]/c} \right. \\
&\quad \left. \times \hat{n}_l^T \cdot \tilde{\mathbf{h}}(f) \cdot \hat{n}_l \right\} \\
&= \int df e^{i2\pi f t} \left\{ -\frac{i\pi f L_l}{c} \text{sinc} \left[\pi f L_l/c (1 - \hat{k} \cdot \hat{n}_l) \right] \right. \\
&\quad \left. \times e^{-i\pi f [L_l + \hat{k} \cdot (\mathbf{r}_s + \mathbf{r}_r)]/c} \hat{n}_l^T \cdot \tilde{\mathbf{h}}(f) \cdot \hat{n}_l \right\} \\
&= \int df e^{i2\pi f t} \left\{ -\frac{i2\pi f L_l}{c} \mathcal{T}_{sr}(f, t) (h_+ \zeta_l^+ + h_\times \zeta_l^\times) \right\}. \tag{87}
\end{aligned}$$

Here $\hat{n}_l \cdot \tilde{\mathbf{h}}(f) \cdot \hat{n}_l = \hat{n}_l \cdot (\tilde{h}_+ \epsilon^+ + \tilde{h}_\times \epsilon^\times) \cdot \hat{n}_l = \tilde{h}_+ \zeta_l^+ + \tilde{h}_\times \zeta_l^\times$. If the GW tensor can be decomposed into $\tilde{\mathbf{h}} = \mathbf{P}(f) \tilde{h}(f)$, where $\mathbf{P} = \epsilon^+ + \epsilon^\times$. Thus, the transfer function for the relative frequency deviation can be written as [40, 41]

$$\begin{aligned}
G_{\text{slr}}^{\text{GW}}(f, t) &= -\frac{i\pi f L_l}{c} \text{sinc} \left[\pi f L_l/c (1 - \hat{k} \cdot \hat{n}_l) \right] \\
&\quad \times e^{-i\pi f [L_l + \hat{k} \cdot (\mathbf{r}_s + \mathbf{r}_r)]/c} \hat{n}_l^T \cdot \mathbf{P}(f) \cdot \hat{n}_l. \tag{88}
\end{aligned}$$

For the multiple modes, the transfer function $G_{\text{slr}}^{\ell m}(f, t)$ has the same form as the above equation, where just $\mathbf{r}$ should be changed to $P_{\ell m}$, i.e.

$$G_{\text{slr}}^{\ell m}(f, t) = -\frac{i\pi f L_l}{c} \text{sinc} \left[\pi f L_l/c (1 - \hat{k} \cdot \hat{n}_l) \right] e^{-i\pi f [L_l + \hat{k} \cdot (\mathbf{r}_s + \mathbf{r}_r)]/c} \hat{n}_l^T \cdot P_{\ell m} \cdot \hat{n}_l. \tag{89}$$

With help of ξ_+ and $\xi_\times$, the part of $\hat{n}_l \cdot P_{\ell m} \cdot \hat{n}_l$ will be

$$\begin{aligned}\hat{n}_l \cdot P_{\ell m} \cdot \hat{n}_l &= \frac{1}{2} {}_{-2}Y_{\ell m}(\ell, \varphi) e^{-i2\psi} (\zeta_l^+ + i\zeta_l^\times) \\ &+ \frac{1}{2} (-1)^\ell {}_{-2}Y_{\ell, -m}^*(\ell, \varphi) e^{+i2\psi} (\xi_l^+ - i\xi_l^\times).\end{aligned}\quad (90)$$

One should note that in the previous equations, when the higher modes are considered, the time–frequency relationship should be considered. With the help of the SPA, the time–frequency relationship will be

$$t_f^{\ell m} = -\frac{1}{2\pi} \frac{d\Psi_{\ell m}}{df}, \quad (91)$$

for different modes.

As for GW with eccentricity, we could not simply calculate the response function using the formulae above, even if it only has the dominant spin-weighted spherical harmonic $(\ell, m) = (2, 2)$. Different eccentric harmonics also have different time–frequency correspondence, so we need to write [11]

$$t_f^j = \frac{1}{2\pi} \frac{d\Psi_j}{df}. \quad (92)$$

Then we decompose $\tilde{\mathbf{h}}$ into eccentric harmonics $\tilde{\mathbf{h}}_j$, i.e.

$$\tilde{\mathbf{h}} = \sum_j \tilde{\mathbf{h}}_j, \quad (93)$$

$$\tilde{\mathbf{h}}_j = P_+ \tilde{h}_j^+ + P_\times \tilde{h}_j^\times, \quad (94)$$

and rewrite equations (87)–(89):

$$\tilde{y}_{\text{slr}} = \sum_j \mathcal{T}_{\text{slr}}^j(f) : \tilde{h}_j, \quad (95)$$

$$\mathcal{T}_{\text{slr}}^j(f) = G_{\text{slr}}\left(f, t_f^j\right), \quad (96)$$

$$G_{\text{slr}}^{\ell m}(f, t) = -\frac{i\pi f L_l}{c} \text{sinc}\left[\pi f L_l / c \left(1 - \hat{k} \cdot \hat{n}_l\right)\right] e^{-i\pi f [L + \hat{k} \cdot (\mathbf{r}_s + \mathbf{r}_r)] / c} \hat{n}_l^T \otimes \hat{n}_l. \quad (97)$$

4.4. Response for the mildly chirping signals

For mildly chirping binary sources that do not contain the Fourier integral, one can assume that the phase of the GW can be approximated as [54].

$$\Phi(\xi) = 2\pi f_0 \xi + \pi \dot{f}_0 \xi^2 + \varphi_0, \quad (98)$$

where $f_0, \dot{f}_0$ and φ_0 are the initial frequency, frequency deviation and phase, respectively. Thus, the instantaneous frequency can be given as

$$\frac{1}{2\pi} \frac{\partial \Phi(\xi)}{\partial t} = \frac{1}{2\pi} \frac{\partial \Phi(\xi)}{\partial \xi} \frac{\partial \xi}{\partial t} = (f_0 + \dot{f}_0 \xi) \left(1 - \hat{k} \cdot \frac{\partial \mathbf{r}(t)}{\partial t}\right). \quad (99)$$

According to the equation, we may assume a fixed frequency at ξ_0 as

$$f_s = f_0 + \dot{f}_0 \xi_0, \quad (100)$$

and the index s denotes the dependency of the approximated frequency on the time of emission ξ_0 . Here, assuming that the frequency of the GW changes very little, i.e. $\dot{f}_0(\xi_L - \xi_0) \ll f_0$. Then

$$\Phi(\xi) \approx \int dt 2\pi f_s \left(1 - \hat{k} \cdot \frac{\partial \mathbf{r}(t)}{\partial t} \right) = \int d\xi 2\pi f_s = 2\pi (f_0 + \dot{f}_0 \xi_0) \xi + C, \quad (101)$$

where C is some integration constant. Meanwhile, the amplitude of the wave also changes little. Then the plane wave can be described as

$$h(\xi) = A(\xi) e^{i2\pi f_s \xi} \approx A(\xi_0) e^{i2\pi f_s \xi_0} e^{i2\pi f_s (\xi - \xi_0)} = h(\xi_0) e^{i2\pi f_s (\xi - \xi_0)}. \quad (102)$$

In this way, the integration of the GW tensor fluctuation will be [54]

$$\begin{aligned} \int_{\xi_0}^{\xi_L} \mathbf{h}(\xi) d\xi &= \mathbf{P} \int h(\xi_0) e^{i2\pi f_s (\xi - \xi_0)} d\xi = \mathbf{P} \frac{1}{i2\pi f_s} (h(\xi_L) - h(\xi_0)) \\ &= \mathbf{P} \frac{1}{i2\pi f_s} h(\xi_0) (e^{i2\pi f_s (\xi_L - \xi_0)} - 1) \\ &= \mathbf{P} \frac{\sin[\pi f_s (\xi_L - \xi_0)]}{\pi f_s} e^{i\pi f_s (\xi_L - \xi_0)} h(\xi_0) \\ &= \mathbf{P} \frac{\sin[\pi f_s (\xi_L - \xi_0)]}{\pi f_s} e^{i\pi f_s (\xi_L + \xi_0)} A(\xi_0) \\ &= \mathbf{P} \frac{(1 - \hat{k} \cdot \hat{n}) L}{c} \text{sinc} \left[\frac{\pi f_s L}{c} (1 - \hat{k} \cdot \hat{n}) \right] \\ &\quad \times e^{-i\pi f_s [L + \hat{k} \cdot (\mathbf{r}_r + \mathbf{r}_s)]/c} A(\xi_0) e^{i2\pi f_s t_r} \\ &= 2\mathbf{P} \frac{(1 - \hat{k} \cdot \hat{n}) L}{c} \mathcal{T}_{sr}(\hat{k}, f_s, t_r) A(\xi_0) e^{i2\pi f_s t_r}, \end{aligned} \quad (103)$$

where $\mathbf{P}$ is the unit tensor matrix of GW. Here we have used

$$\xi_L - \xi_0 = (t_r - \hat{k} \cdot \mathbf{r}_r) - (t_s - \hat{k} \cdot \mathbf{r}_s) = (1 - \hat{k} \cdot \hat{n}) L/c, \quad (104)$$

$$\begin{aligned} \xi_L + \xi_0 &= (t_r - \hat{k} \cdot \mathbf{r}_r/c) + (t_s - \hat{k} \cdot \mathbf{r}_s/c) \\ &\approx 2t_r - L/c - \hat{k} \cdot (\mathbf{r}_s + \mathbf{r}_r)/c. \end{aligned} \quad (105)$$

If the amplitude of GW is some constant, then the path length variation defined in equation (65) will be

$$\frac{\delta l_{sr}}{L}(t) \approx \mathcal{T}_{sr}(\hat{k}, f_s, t) \hat{n} \cdot \mathbf{h}(t) \cdot \hat{n}. \quad (106)$$

And according to equation (70), one can find that

$$\frac{\delta \nu}{\nu_0} = -\frac{i2\pi f_s L}{c} \frac{\delta l}{L}. \quad (107)$$

This is similar to the response in the frequency domain, and one should note that the above formula is valid only when the GW is some mildly chirping signals or some monochromatic signals.

4.5. Response to the Stochastic backgrounds

There are many GW events in the Universe. And the superposition of weak GWs can form an SGWB. It can be expanded in terms of plane waves

$$h_{ij}(t, \mathbf{r}) = \sum_{+, \times} \int \hat{\Omega}_{\hat{k}} \int d\tilde{f} \tilde{h}_{+, \times}(f, \hat{k}) \epsilon^{+, \times}(\hat{k}) : \mathbf{D}(\hat{k}, f) e^{-i2\pi f(t - \hat{k} \cdot \mathbf{r}/c)}. \quad (108)$$

where $\int d\hat{\Omega} = \int_0^{2\pi} d\phi \int_0^\pi \sin\theta d\theta$ denotes an integral in all the sky, $\mathbf{r}$ is the location of the measurement at time t .

The detector response to the superposition of GW can be written as the convolution of the metric perturbations $h_{ij}(t, \mathbf{r})$ with the impulse response $F^{ij}(t, \mathbf{r})$ of the detector

$$\begin{aligned} s(t) &= (\mathbf{R} \star \mathbf{h})(t, \hat{r}) \equiv \int d\tau \int d^3x F^{ij}(\tau, \mathbf{x}) h_{ij}(t - \tau, \mathbf{r} - \mathbf{x}) \\ &= \int df \int d\hat{\Omega}_{\hat{k}} R^{ij}(f, \hat{k}) h_{ij}(f, \hat{k}) e^{-i2\pi ft} = \int df \tilde{s}(f) e^{-i2\pi ft}, \end{aligned} \quad (109)$$

where

$$R^{ij}(f, \hat{k}) = e^{i2\pi f \hat{k} \cdot \mathbf{r}/c} \int d\tau \int d^3x F^{ij}(\tau, \mathbf{x}) e^{-i2\pi f(\tau + \hat{k} \cdot \mathbf{x}/c)}. \quad (110)$$

is the response function in the frequency domain, and $\tilde{s}(f)$ is the signal in frequency domain.

For individual detectors, the analysis focuses on the response function of a single channel, particularly the orthogonal TDI channels such as the A/E channels. However, when considering detector networks, the correlation between different channels becomes a critical factor. In this context, the response function generalizes to the overlap reduction function (ORF) for distinct channels. Notably, the ORF exhibits a time dependence as a result of the orbital motion of the detectors. To account for this variability, one can define the time-averaged ORF over the total correlation time T_{tot} as follows [14]:

$$\bar{\Gamma}_{ij}(f) = \sqrt{\frac{1}{T_{\text{tot}}} \int_0^{T_{\text{tot}}} dt |\hat{\Gamma}_{ij}(f, t_0)|^2}. \quad (111)$$

For networks such as TianQin I+II or TianQin + LISA, four distinct channel pairs can be utilized for cross-correlation analysis. Consequently, the total ORF for these networks is defined as the quadrature sum of the individual time-averaged ORFs [55]

$$\Gamma_{\text{tot}}(f) = \sqrt{\sum_{I, J} |\bar{\Gamma}_{IJ}(f)|^2}, \quad (112)$$

where I and J label a pair of orthogonal channels.

5. Time delay interference

The signal transmitted from spacecraft S_s that is received at spacecraft S_r at time t_r has its phase compared to the local reference to give the output of the phase change $\Phi_{\text{sr}}(t_r)$ [54]. The phase difference has contributions from the laser phase noise $C(t)$, optical path length variations, shot noise $n^s(t)$ and acceleration noise $\mathbf{n}^a(t)$ [54]

$$\begin{aligned} \Phi_{\text{sr}}(t_r) &= C_s(t_s) - C_r(t_r) + 2\pi\nu_0(\delta l_l(t_s) + \Delta l_l(t_s)) \\ &\quad + n_{\text{sr}}^s(t_r) - \hat{n}_l(t_s) \cdot [\mathbf{n}_{\text{sr}}^a(t_r) - \mathbf{n}_{\text{rs}}^a(t_s)], \end{aligned} \quad (113)$$

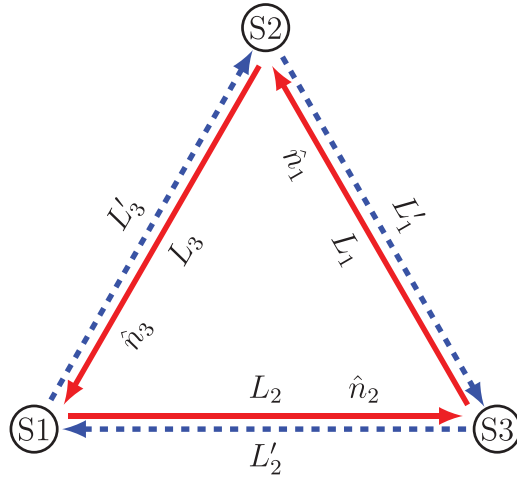


Figure 8. Illustration of detector constellation. Three satellites are marked as 1, 2, and 3. Laser paths are marked as L_i and L'_i , where L'_i represents the direction opposite to L_i . The direction of unit vector $\hat{n}_i$ is the same as that of L_i .

where t_s is given implicitly by $t_s = t_r - \ell_{sr}(t_s)$ and ν_0 is the laser frequency. The optical path length variations caused by GWs are $\delta l_i(t_s)$, and those caused by orbital effects is $\Delta l_i(t_s)$. From equation (113), one can find that the space-based GW detection suffers from laser phase noise, which can be alleviated through TDI technology. TDI involves heterodyne interferometry with unequal arm lengths and independent phase-difference readouts [56]. By essentially constructing a virtually equal-arm interferometer, the laser phase noise cancels out exactly.

5.1. General TDI combination

Before introducing the TDI, let us first introduce some definitions. In figure 8, the satellite numbers in space are defined clockwise. The definition of the laser path counterclockwise is the positive direction ($\hat{n}_i$), denoted as L_i , and clockwise is the negative direction ($-\hat{n}_i$), denoted as L'_i . The arm length $|L_i|$ is defined as the distance between the other two satellites facing the satellite i , where $i = 1, 2, 3$.

As shown in figure 8, let $\vec{X}_i$ as the i th spacecraft, l_{ij} is the distance between the i th and j th spacecrafts, then

$$\begin{aligned} L_1 &= \vec{u}_{32} l_{32} = \vec{X}_2 - \vec{X}_3, L_2 = \vec{u}_{13} l_{13} = \vec{X}_3 - \vec{X}_1, \\ L_3 &= \vec{u}_{21} l_{21} = \vec{X}_1 - \vec{X}_2. \end{aligned} \quad (114)$$

$$\begin{aligned} \hat{n}_1 &= \vec{u}_{32} = \frac{\vec{X}_2 - \vec{X}_3}{|\vec{X}_2 - \vec{X}_3|}, \hat{n}_2 = \vec{u}_{13} = \frac{\vec{X}_3 - \vec{X}_1}{|\vec{X}_3 - \vec{X}_1|}, \\ \hat{n}_3 &= \vec{u}_{21} = \frac{\vec{X}_1 - \vec{X}_2}{|\vec{X}_1 - \vec{X}_2|} \end{aligned} \quad (115)$$

Let s_1 as the time-dependent phase change signals received by the 1st spacecraft, which is sent from the 2nd spacecraft and propagates along the link L_3 . One can also sign it as s_{231} . Similarly,

let s'_1 as the signal received by spacecraft 1, which is sent from spacecraft 3 and propagates along L'_2 , or recorded as s_{321} . As shown in figure 8, there are six independent laser links.

The first generation TDI combination does not consider the rotation and flexing of the spacecraft constellation, which is only valid for a static constellation, i.e.

$$L_i(t) = L_i = \text{const}, \quad L_i = L_{i'}. \quad (116)$$

This means that all the arm lengths remain constant as time evolves, and the time duration of photon propagation along the arm is independent of the direction of the photons. The 1.5 or modified TDI generation is valid for a rigid but rotating spacecraft constellation, i.e.

$$L_i(t) = L_i = \text{const}, \quad L_i \neq L_{i'}. \quad (117)$$

The propagation direction of photons should be considered. The second generation TDI combination is applied to consider a rotating and flexing constellation, i.e.

$$L_i(t) = L_i + \dot{L}_i t, \quad L_i \neq L_{i'}. \quad (118)$$

The arm length changes linearly in time, and relative to the velocity of $\dot{L}_i$. Here, define the time delay operator as $\mathcal{D}_i$, where

$$\mathcal{D}_i x(t) \equiv x(t - L_i/c), \quad (119)$$

$$\mathcal{D}_i \mathcal{D}_j x(t) = \mathcal{D}_{ij} x(t) \equiv x(t - L_i/c - L_j/c). \quad (120)$$

Then one can define the 1.5 generation unequal arm Michelson-like combination as (see figure 9) [50]

$$\begin{aligned} X_{1.5} &= y_{32'1} + \mathcal{D}_{2'} [y_{123} + \mathcal{D}_2 (y_{231} + \mathcal{D}_3 y_{13'2})] \\ &\quad - y_{231} - \mathcal{D}_3 [y_{13'2} + \mathcal{D}_{3'} (y_{32'1} + \mathcal{D}_{2'} y_{123})] \\ &= y_{32'1} + \mathcal{D}_{2'} y_{123} + \mathcal{D}_{2'2} y_{231} + \mathcal{D}_{2'23} y_{13'2} \\ &\quad - y_{231} - \mathcal{D}_3 y_{13'2} - \mathcal{D}_{33'} y_{32'1} - \mathcal{D}_{33'2'} y_{123}. \end{aligned} \quad (121)$$

For the generation 2.0, one has [50]

$$\begin{aligned} X_{2.0} &= y_{32'1} + \mathcal{D}_{2'} y_{123} + \mathcal{D}_{2'2} y_{231} + \mathcal{D}_{2'23} y_{13'2} + \mathcal{D}_{2'233'} y_{231} \\ &\quad + \mathcal{D}_{2'233'3} y_{13'2} + \mathcal{D}_{2'233'33'} y_{32'1} + \mathcal{D}_{2'233'33'2'} y_{123} - y_{231} \\ &\quad - \mathcal{D}_3 y_{13'2} - \mathcal{D}_{33'} y_{32'1} - \mathcal{D}_{33'2'} y_{123} - \mathcal{D}_{33'2'2} y_{32'1} \\ &\quad - \mathcal{D}_{33'2'22'} y_{123} - \mathcal{D}_{33'2'22'2} y_{231} - \mathcal{D}_{33'2'22'23} y_{13'2}. \end{aligned} \quad (122)$$

The Y and Z channels can be generated by cyclic permutation of indices: $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$.

Suppose that all the armlengths are equal, i.e. $L_i = L$. Thus, in the time domain, the first generation of TDI Michelson-like X channel will be

$$X_{1.0} = [y_{32'1} + \mathcal{D} y_{123}] + \mathcal{D}^2 [y_{231} + \mathcal{D} y_{13'2}] - [y_{231} + \mathcal{D} y_{13'2}] - \mathcal{D}^2 [y_{32'1} + \mathcal{D} y_{123}], \quad (123)$$

where $\mathcal{D} = \mathcal{D}_i$ and $\mathcal{D}^2 = \mathcal{D}\mathcal{D}$. Simply, let $y_{slr,nL} = y_{sr}(t - nL)$, its Fourier transform will be $\tilde{y}_{slr,nL} = \mathcal{D}^n \tilde{y}_{sr}$, where $\mathcal{D}$ is the time delay. Otherwise, one can easily get the frequency domain TDI channel as

$$\begin{aligned} \tilde{X}_{1.0} &= [\tilde{y}_{31} + \mathcal{D} \tilde{y}_{13}] + \mathcal{D}^2 [\tilde{y}_{21} + \mathcal{D} \tilde{y}_{12}] - [\tilde{y}_{21} + \mathcal{D} \tilde{y}_{12}] - \mathcal{D}^2 [\tilde{y}_{31} + \mathcal{D} \tilde{y}_{13}] \\ &= (1 - \mathcal{D}^2) [\tilde{y}_{31} + \mathcal{D} \tilde{y}_{13} - \tilde{y}_{21} - \mathcal{D} \tilde{y}_{12}], \end{aligned} \quad (124)$$

Similarly, the second generation of TDI Michelson-like X channel (as shown in figure 10) in the frequency domain will be

$$\tilde{X}_{2.0} = \tilde{y}_{31} + \mathcal{D} \tilde{y}_{13} + \mathcal{D}^2 \tilde{y}_{21} + \mathcal{D}^3 \tilde{y}_{12} + \mathcal{D}^4 \tilde{y}_{21} + \mathcal{D}^5 \tilde{y}_{12} + \mathcal{D}^6 \tilde{y}_{31} + \mathcal{D}^7 \tilde{y}_{13}$$

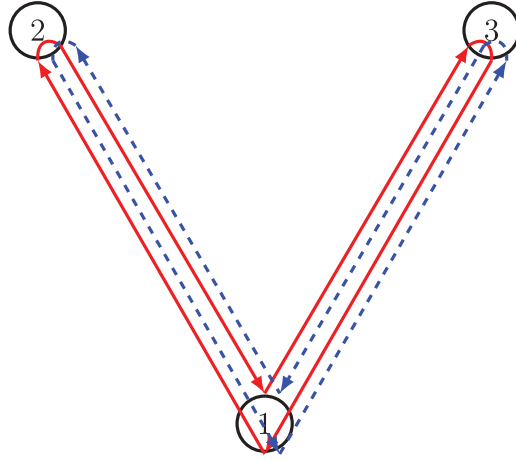


Figure 9. Michelson-like TDI-X channel of first generation TDI.

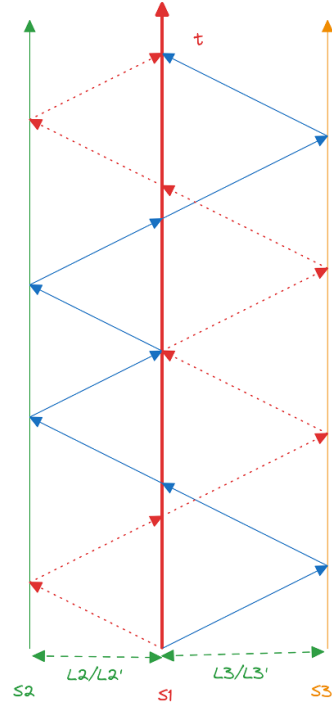


Figure 10. Space-time map of TDI 2.0 for the Michelson-like X channel.

$$\begin{aligned}
 & -\tilde{y}_{21} - \mathcal{D}\tilde{y}_{12} - \mathcal{D}^2\tilde{y}_{31} - \mathcal{D}^3\tilde{y}_{13} - \mathcal{D}^4\tilde{y}_{31} - \mathcal{D}^5\tilde{y}_{13} - \mathcal{D}^6\tilde{y}_{21} - \mathcal{D}^7\tilde{y}_{12} \\
 & = (1 - \mathcal{D}^4)\tilde{X}_{1,0}.
 \end{aligned} \tag{125}$$

However, different channels will use the same link, then the instrumental noises in different channels may be correlated with each other. Considering that all the satellites are identical, we

can get one ‘optimal’ combination by linear combinations of X , Y , and Z [57]:

$$A = \frac{1}{\sqrt{2}} (Z - X), \quad (126)$$

$$E = \frac{1}{\sqrt{6}} (X - 2Y + Z), \quad (127)$$

$$T = \frac{1}{\sqrt{3}} (X + Y + Z). \quad (128)$$

In the A , E , and T channels, the instrumental noise is orthogonal, and consequently, the noise correlation matrix of these three combinations is diagonal [57].

5.2. Instrument noise

We will focus on the case where the instrumental noise $n(t)$ is assumed to be Gaussian stationary with a zero mean. Thus, the ensemble average of the Fourier components of the noise $n(f)$ can be written in the following form

$$\langle \tilde{n}(f) \tilde{n}^*(f') \rangle = \frac{1}{2} \delta(f - f') S_n(f), \quad (129)$$

where $*$ denotes complex conjugate, and $S_n(f)$ is the single-sided noise power spectra density (PSD)⁹.

For TianQin, the designed requirement for the acceleration noise is $\sqrt{S_a} = 10^{-15} \text{ms}^{-2} \text{Hz}^{-1/2}$ and the displacement noise is $\sqrt{S_x} = 1 \text{pmHz}^{-1/2}$ [1]. For LISA, as reported in reference [58], the displacement noise is $\sqrt{S_x} = 15 \text{pmHz}^{-1/2}$ and for the acceleration noise is $\sqrt{S_a} = 3 \times 10^{-15} \text{ms}^{-2} \text{Hz}^{-1/2}$. For Taiji, the design goal for the displacement noise is $\sqrt{S_x} = 8 \text{pmHz}^{-1/2}$ and for the acceleration noise is $\sqrt{S_a} = 3 \times 10^{-15} \text{ms}^{-2} \text{Hz}^{-1/2}$ at 1 mHz [29].

As discussed at the beginning of section 5, when the laser noise is canceled, the total can be described by two noises. One is displacement or position noise, which is dominated at high frequencies. The other one is the acceleration noise, which is dominated at low frequencies. Note that the noise parameters defined in the previous paragraph should convert to the same dimension, such as in the dimension of length (here, using the LISA noise as an example)

$$\sqrt{S_{\delta l}^{\text{oms}}}(f) = \sqrt{S_x} \sqrt{1 + \left(\frac{2\text{mHz}}{f} \right)^4}, \quad (130)$$

$$\sqrt{S_{\delta l}^{\text{acc}}}(f) = \frac{\sqrt{S_a}}{(2\pi f)^2} \sqrt{1 + \left(\frac{0.4\text{mHz}}{f} \right)^2} \sqrt{1 + \left(\frac{f}{8\text{mHz}} \right)}. \quad (131)$$

and in the dimension of the relative frequency, it will be

$$\sqrt{S_{\delta \nu/\nu}^{\text{oms}}}(f) = \sqrt{S_x} \frac{2\pi f}{c} \sqrt{1 + \left(\frac{2\text{mHz}}{f} \right)^4}, \quad (132)$$

$$\sqrt{S_{\delta \nu/\nu}^{\text{acc}}}(f) = \frac{\sqrt{S_a}}{2\pi f c} \sqrt{1 + \left(\frac{0.4\text{mHz}}{f} \right)^2} \sqrt{1 + \left(\frac{f}{8\text{mHz}} \right)}. \quad (133)$$

⁹ Because $n(t)$ is real, $\tilde{n}^*(f) = \tilde{n}(-f)$ and therefore $S_n(-f) = S_n(f)$.

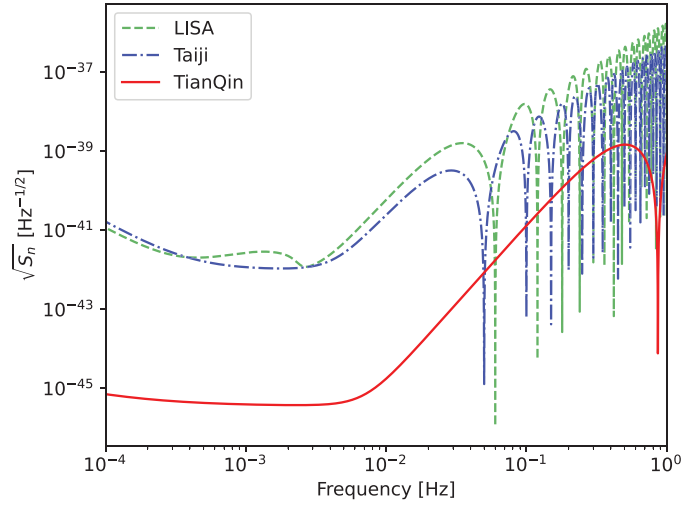


Figure 11. Noise PSD of TDI A channel for LISA, Taiji, and TianQin (four year data).

For different detectors, the difference is the value at the front and the tail of the frequency variation. For TianQin, the relative noise parameters will be [1]

$$\sqrt{S_{\delta l}^{\text{oms}}}(f) = \sqrt{S_x}, \quad \sqrt{S_{\delta l}^{\text{acc}}}(f) = \frac{\sqrt{S_a}}{(2\pi f)^2} \sqrt{1 + \frac{0.1\text{mHz}}{f}}, \quad (134)$$

$$\sqrt{S_{\delta\nu/\nu}^{\text{oms}}}(f) = \sqrt{S_x} \frac{2\pi f}{c}, \quad \sqrt{S_{\delta\nu/\nu}^{\text{acc}}}(f) = \frac{\sqrt{S_a}}{2\pi f c} \sqrt{1 + \frac{0.1\text{mHz}}{f}}. \quad (135)$$

With the above definitions and the assumption that all the instrument's noise parameters are the same, the PSD and cross-spectrum of noise for the TDI-1.0 in the X, Y, Z channels will be [50, 58]

$$S_n^{X,1.0} = 16 \sin^2(f/f_*) [S^{\text{oms}} + 2(1 + \cos^2(f/f_*)) S^{\text{acc}}], \quad (136)$$

$$S_n^{XY,1.0} = -8 \sin^2(f/f_*) \cos(f/f_*) (S^{\text{oms}} + 4S^{\text{acc}}), \quad (137)$$

where $f_* = c/(2\pi L)$. For the A, E, T channels, the PSDs will be [50, 58]

$$S_n^{A,E}(f) = 8 \sin^2(f/f_*) \{4[1 + \cos(f/f_*) + \cos^2(f/f_*)] S^{\text{acc}} + [2 + \cos(f/f_*)] S^{\text{oms}}\}, \quad (138)$$

$$S_n^T(f) = 32 \sin^2(f/f_*) \sin^2(f/f_*/2) [4 \sin^2(f/f_*/2) S^{\text{acc}} + S^{\text{oms}}]. \quad (139)$$

And the noise PSD of the second-generation TDI channels is

$$S_n^{\mathcal{I},2.0} = 4 \sin^2(2f/f_*) S_n^{\mathcal{I},1.0} \quad (140)$$

where $\mathcal{I} = X, Y, Z, XY, \dots, A, E, T$.

In figure 11, we have shown the three noise PSD curves of TDI first generation A channel for LISA, Taiji, and TianQin.

6. Waveform

To extract information from the detector data, one should model the entire detection process. With the basic definition in section 4.1, one can build a model for some general GW signals. To know the type of GW source and more information about the GW systems, an exact waveform is needed. In this section, we review some waveforms we use for each type of source.

6.1. GCB

In the mHz frequency band, GW events are mainly composed of white dwarf binaries (WDBs) in the Milky Way (with the number $\sim \mathcal{O}(10^8)$) [59], which are expected to be the most numerous GW sources for SBD. These GCBs are expected to exhibit relatively little frequency evolution. Thus, the GW strain emitted from a GCB can be safely approximated as (in the source frame) [51]

$$h_+(t) = A_+ \cos \Phi(t) = h_0 \frac{1 + \cos^2 \iota}{2} \cos \Phi(t), \quad (141)$$

$$h_\times(t) = A_\times \sin \Phi(t) = h_0 \cos \iota \sin \Phi(t), \quad (142)$$

$$h_0 = \frac{4(G\mathcal{M}_c)^{5/3}}{c^4 D_L} (\pi f)^{2/3}, \quad (143)$$

$$\Phi(t) = 2\pi f t + \pi \dot{f} t^2 + \frac{\pi}{3} \ddot{f} t^3 + \phi_0, \quad (144)$$

where ι is the inclination angle of the quadruple rotation axis with respect to the line of sight (here the direction is from the source to the Sun), $\mathcal{M}_c = (m_1 m_2)^{3/5} / (m_1 + m_2)^{1/5}$ is the chirp mass of the system (m_1 and m_2 are the individual masses of the components of the binary), D_L is the luminosity distance to the source, ϕ_0 is the initial phase at the start of the observation, f , $\dot{f}$, and $\ddot{f}$ are the frequency of the source, the frequency's derivative, and the double derivative with respect to time, and $\ddot{f} = \frac{11}{3} \frac{\dot{f}^2}{f^2}$.

Considering the motion of the detectors moving around the Sun, a Doppler modulation of the phase of the waveform should be taken into account, i.e.

$$\Phi(t) \rightarrow \Phi(t) + \Phi_D(t), \quad (145)$$

$$\Phi_D(t) = 2\pi (f + \dot{f}t) \frac{R}{c} \cos \beta \cos(2\pi f_m t - \lambda), \quad (146)$$

where $\Phi_D(t)$ is the Doppler modulation, $f_m = 1/\text{year}$ is the modulation frequency, β and λ are the latitudes and the longitude of the source in ecliptic coordinates, $R = 1\text{AU}$ is the semi-major axis of the guiding center of the satellite constellation, respectively. However, one should note that the Doppler modulation shown in equation (146) needn't be considered if we use the response defined in the SSB coordinate and the response function defined in section 4.4. Because $\mathbf{k} \cdot \mathbf{p}_0$ will be the Doppler effect.

6.2. Black hole binary (BHB)

General phenomenological waveform For a BHB system, one can describe its waveform in the time domain or frequency domain with the help of the SPA. Here, we consider the frequency domain IMRPhenomD waveform, which assumes aligned spin, so only two parameters are needed to describe the spin parameters [60, 61]. In this frame, a BHB system can be characterized by four intrinsic parameters: masses (m_1, m_2) and dimensionless spins (χ_1, χ_2); seven

extrinsic parameters: luminosity distance D_L , inclination angle ι , polarization angle ψ , coalescence time and phase (t_c, ϕ_c) and the ecliptic longitude and ecliptic latitude (λ, β) in the SSB. In the IMRPhenomD waveform model, the waveforms of plus and cross modes will be

$$\tilde{h}_+(f) = \frac{\mathcal{M}_c^{5/6}}{\pi^{2/3} D_L} \frac{1 + \cos^2 \iota}{2} f^{-7/6} \exp(i\Psi(f)), \quad (147)$$

$$\tilde{h}_\times(f) = -i \frac{\mathcal{M}_c^{5/6}}{\pi^{2/3} D_L} \cos \iota f^{-7/6} \exp(i\Psi(f)). \quad (148)$$

More details about the phase $\Psi(f)$ can be seen in [60].

Eccentric waveform The GW emission causes the circularization effect, which makes the binaries almost non-eccentric when they are in the GBD frequency band. But when the binaries are in the SBD frequency band, the eccentricity should be taken into account. Many eccentric waveform models have been developed to date [62]. Here we use *EccentricFD*, which is a frequency-domain third post-Newtonian (3PN) waveform with initial eccentricity e_0 valid up to 0.4 [48, 63], and has been included in *LALSuite* [64]. This analytic model only contains the inspiral process of a binary, however, it is sufficient for SBHBs, as they are likely to merge outside the sensitive frequency band of SBDs.

Note that the BHB system can be divided into MBHB and SBHB systems according to their masses and origin. The heavier BHB systems have lower frequency bands. Though their origin or characteristics are different, their waveform formulas are similar. When analyzing the data, it is important to note the range of parameter values and the applicability of the waveform.

6.3. Extreme mass ratio inspirals (EMRIs)

A prominent source that space-based detectors will detect is EMRIs. These sources are formed by a stellar-mass compact object such as a stellar origin black hole with a mass $\mu \sim 1 - 10^2 M_\odot$ inspiraling into a massive black hole (MBH) with a mass $M \sim 10^4 - 10^7 M_\odot$ [12, 65]. Specifically, the time domain dimensionless strain of an EMRI source $h(t)$ can be given by

$$h(t) = \frac{\mu}{d_L} \sum_{lmkn} A_{lmkn}(t) S_{lmkn}(t, \theta) e^{im\phi} e^{-i\Phi_{mkn}(t)}, \quad (149)$$

where t is the time of arrival of the GW at the SSB, θ is the source-frame polar viewing angle, ϕ is the source-frame azimuthal viewing angle, d_L is the luminosity distance, and $\{l, m, k, n\}$ are the indices describing the frequency-domain harmonic mode decomposition [66, 67]. The indices l, m, k , and n label the orbital angular momentum, azimuthal, polar, and radial modes, respectively. $\Phi_{mkn} = m\Phi_\varphi + k\Phi_\theta + n\Phi_r$ is the summation of decomposed phases for each given mode. The amplitude A_{lmkn} is related to the amplitude Z_{lmkn}^∞ of the Teukolsky mode amplitude far from the source. It is given by $A_{lmkn} = -2Z_{lmkn}^\infty / \omega_{mkn}^2$, where $\omega_{mkn} = m\Omega_\varphi + k\Omega_\theta + n\Omega_r$ is the frequency of the mode, and $\Omega_{r, \theta, \phi}$ describe the frequencies of a Kerr geodesic orbit.

To expedite the generation of EMRI signals, we utilize the *FastEMRIWaveform* (FEW) package¹⁰. The FEW is optimized to generate GW signals efficiently with GPU acceleration [67, 68]. The trajectory calculation is done through a large-scale cubic-spline interpolation of $A_{lmn}(t)$ and $\Phi_{r, \varphi}(t)$ at a sparse set of times ($\sim 10^2$). A reduced-order-model technique is

¹⁰ <https://github.com/BlackHolePerturbationToolkit/FastEMRIWaveforms>.

employed in actuality, reducing the number of harmonic modes needed by approximately 40 times and thereby significantly cutting down the time needed to generate the waveform for each source [66, 68]. For example, $l \in [2, 10]$, $m \in [0, l]$ and $n \in [-30, 30]$, which totals 3843 modes reduce to $\sim 10^2$ modes. So far, the fully relativistic FEW model is limited to eccentric orbits in the Schwarzschild spacetime.

6.4. Stochastic GWs background

In addition to the aforementioned primary distinguishable GW sources, there is another important type of GW source that could potentially be detected by SBD, known as the SGWB. SGWB is composed by a huge number of independent and unresolved GW sources [53]. These stochastic signals are effectively another source of noise in GW detectors. A SGWB can be written as a superposition of plane waves with frequencies of f and coming from different directions $\hat{k}$ on the sky

$$h_{ij}(t, \mathbf{x}) = \sum_P \int_{-\infty}^{+\infty} df \int_{S^2} d\Omega_{\hat{k}} \tilde{h}_P(f, \hat{k}) e_{ij}^P(\hat{k}) e^{i2\pi f[t - \hat{k} \cdot \mathbf{x}(t)/c]}, \quad (150)$$

where $P = \{+, \times\}$ denotes polarization. As a stochastic source, one can treat the complex amplitude $\tilde{h}_P(f, \hat{k})$ as some random variable with zero mean value. Supposing the SGWB is stationary, Gaussian, isotropic, and unpolarized, the ensemble average of the two random amplitudes $\tilde{h}_P(f, k)$ can be defined as [53, 69]

$$\langle \tilde{h}_P(f, \hat{k}) \tilde{h}_{P'}^*(f', \hat{k}') \rangle = \delta(f - f') \frac{\delta^2(\hat{k}, \hat{k}')}{4\pi} \delta_{PP'} \frac{1}{2} S_h(f). \quad (151)$$

The function $S_h(f)$ is the one-sided PSD of SGWB.

Note here that $\delta^2(\hat{k}, \hat{k}')$ is a Dirac delta over the two-sphere, and it implies that the SGWB is independent of $\hat{k}$. However, it is expected that the SBDs will detect millions of WDBs in the Milky Way and nearby Universe [16], and the superposition of millions of unresolved WDBs will contribute to an SGWB [70] (often referred to as foreground due to its strength). Furthermore, due to our location at one end of the Milky Way, this SGWB is anisotropic. Of course, there may exist other anisotropic SGWBs as well [20]. In this case, the PSD of the anisotropic SGWB will depend on the frequency and direction as $\mathcal{P}(f, \hat{k})$. If we assume SGWB is directional and frequency independent, the PSD can be factorized as [71]

$$\mathcal{P}(f, \hat{k}) = H(f) \mathcal{P}_h(\hat{k}) \quad (152)$$

where the PSD of the SGWB is given by $H(f)$, and the $\mathcal{P}_h(\hat{k})$ describes the distribution of signal.

7. Example data-set

To simulate the joint observation of certain GW signals, it is necessary to have precise knowledge of the relative positions of the three detectors. The relative positions of guiding centers for each detector can be determined by the initial phase parameter $\alpha - \beta$ or κ_0 and α'' as defined in equations (19)–(21) and (25)–(27). Additionally, the relative position of the spacecrafts in different detectors can be determined by the initial phase of the spacecraft (here is

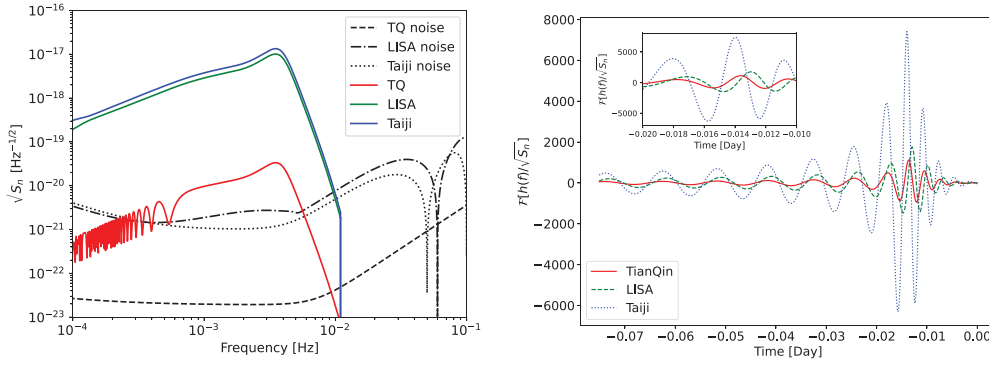


Figure 12. Left: TDI-A channel responded MBHB signal and relative noise PSD detected by different detectors in frequency domain. Right: the signals in time domain, and the signal here is the strain divided by the noise ASD, which can make the responded signals within similar magnitude. The masses for the binary system are $(3.5 \times 10^6, 2.1 \times 10^5) M_\odot$, spins are $(0.2, 0.1)$, the luminosity distance is 10^3 Mpc, the position is $(\lambda, \beta) = (0.4, 1.2)$, and $\iota = 0.3, t_c = 0$. The total observation time is three months. Here, the IMRPhenomD waveform is applied. The initial phase of TianQin's and LISA's first spacecraft is set to 0 for this figure.

the initial phase parameter λ and λ'). Once the detectors are launched, the relative phase and positions are fixed. However, when simulating data for testing purposes, the initial phase parameters are some arbitrary values.

MBHB are the primary sources for SBDs, and its total inspiral-merger-ringdown phase can be detected in the mHz band. In figure 12, we have shown the MBHB event detected by TianQin, LISA, and Taiji and relative noise PSD. From the left figure, it can be found that the length of the arms gives LISA and Taiji an advantage in terms of the response intensity to signals, but at the same time, it also results in higher low-frequency noise levels. From the right panel, one can find that the amplitude and the phase of the response signal of the three detectors varies. From the different phases, one can say that the signal will arrival at different detectors at different time. This is the key feature of the joint simulating of GW signals. The time delay between the arrival of the signals from different detectors can be calculated by the relative positions of the detectors and the relative positions of the source.

The mass of SBHB systems is relatively lighter compared to MBHB, which leads to these systems predominantly producing signals in higher frequency ranges. In the figure 13, we can observe the performance of a SBHB signal across different detectors. Interestingly, when eccentricity is taken into account, the response waveform becomes considerably more intricate compared to the case where eccentricity is disregarded. This increased complexity in the waveform poses significant challenges for data processing and analysis. Furthermore, the figure demonstrates that the intersection point between the curve of the response signal from TianQin and the noise PSD is noticeably higher than in the case of LISA or Taiji. This observation suggests that TianQin exhibits certain advantages in high-frequency detection.

In the low-frequency region of figure 12, the post-response signal of TianQin shows oscillations. Likewise, in figure 13, the response signals from all three detectors demonstrate oscillatory behavior. These oscillations arise as a result of the orbital motion of the detectors.

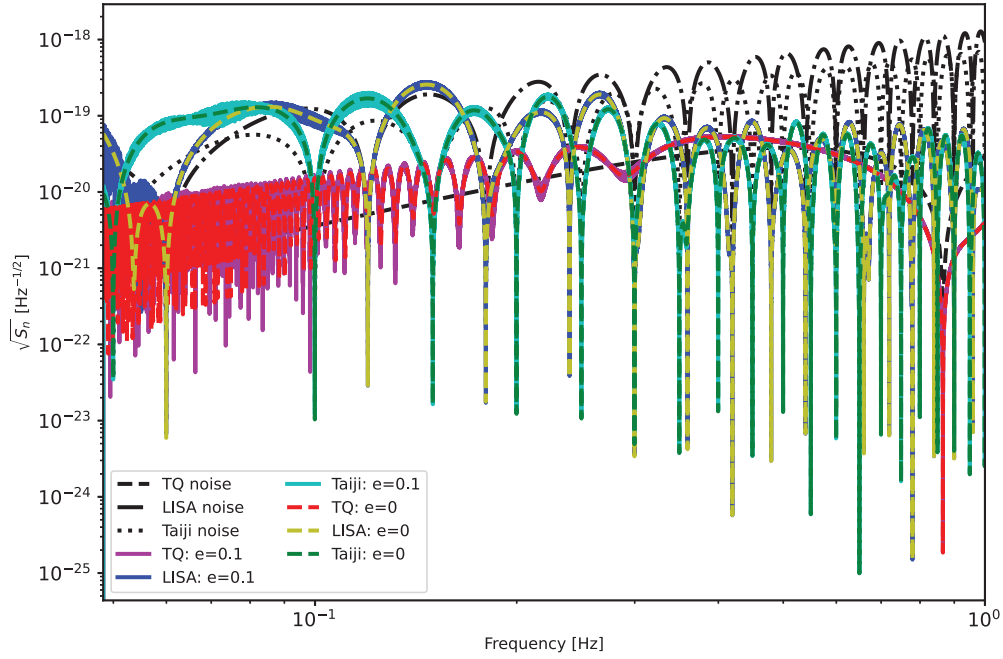


Figure 13. TDI-A channel responded SBHB signal and relative noise PSD detected by different detectors. The masses for the binary system are $(35.6, 30.6) M_{\odot}$, the luminosity distance is 100 Mpc, the position is $(\lambda, \beta) = (4.7, -1.5)$, and $\iota = 0.3, t_c = 0$. The total observation time is three months. Here we have used the *EccentricFD* waveform. The initial phase of TianQin's and LISA's first spacecraft is set to 0 for this figure.

8. Summary

Around 2035, one may see more than one SBDs operating simultaneously, with potential candidates including TianQin, LISA, and Taiji. Apart from the huge prospect of the scientific return from the joint observation over single detectors [36], there are also challenges in doing data analysis for joint observation. To facilitate the study of problems involved in the joint data analysis, we have introduced *GWSpace* in this paper, which is a package that can simulate the joint detection data from three SBDs: TianQin, LISA, and Taiji.

GWSpace uses SSB as the common coordinate system for all detectors. It can simulate data for GCB, BHB, EMRI, SGWB, and simple burst signals. It supports injecting time-domain waveform functions and obtaining observed data through time-domain responses. For frequency-domain waveforms, it supports the frequency-domain responses of regular 22 mode, higher harmonic modes, and waveforms with eccentricity BHB. The first TDI combinations are now included in the time domain and the frequency domain. It includes the time-domain and frequency-domain responses of the 1st generation TDI combinations and the corresponding TDI noise. We have also given some example data sets generated with the package. The package is open source and is free for downloading and using¹¹. To clearly define all the notations and to eliminate possible misunderstanding, we have presented a detailed description of

¹¹ <https://github.com/TianQinSYSU/GWSpace>.

the coordinate system, the detector orbits, the detector responses, the TDI combinations, the instrumental noise models, and the waveforms for each source in this paper.

As the first work in this direction, GWSpace can be further improved in many ways. For example, we have only implemented the first generation TDI so far, while second-generation combinations are usually required, at least for LISA and Taiji. Moreover, a more robust response is needed for some sources with complex waveforms, such as BHB systems with eccentricity [11]. The package still relies on very idealistic assumptions about the noise: the noises from all satellites in a detector are identical, while in reality, no two spacecraft can be the same [15].

One can improve on the last point by implementing more sophisticated noise models for each detector, but the most precise noise model will have to come from the people responsible for each detector. We are hopeful that this may happen one day, and then GWSpace can serve as the starting point for a serious multi-mission data challenge for space-based GW detection.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://github.com/TianQinSYSU/GWSpace>.

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References

- [1] Luo J *et al* 2016 *Class. Quantum Grav.* **33** 035010
- [2] Team K 1997 *Class. Quantum Grav.* **14** 1399–404
- [3] Amaro-Seoane P *et al* 2017 Laser interferometer space antenna (arXiv:1702.00786)

¹² <https://excalidraw.com/>.

- [4] Gong X *et al* 2015 *J. Phys.: Conf. Ser.* **610** 012011
- [5] Hu W-R and Wu Y-L 2017 *Natl Sci. Rev.* **4** 685–6
- [6] Abbott R *et al* (The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration) 2021 arXiv:2111.03606
- [7] Huang S-J, Hu Y-M, Korol V, Li P-C, Liang Z-C, Lu Y, Wang H-T, Yu S and Mei J 2020 *Phys. Rev. D* **102** 063021
- [8] Lu Y, Li E-K, Hu Y-M, Zhang J-D and Mei J 2023 *Res. Astron. Astrophys.* **23** 015022
- [9] Wang H-T *et al* 2019 *Phys. Rev. D* **100** 043003
- [10] Liu S, Hu Y-M, Zhang J-D and Mei J 2020 *Phys. Rev. D* **101** 103027
- [11] Wang H, Harry I, Nitz A and Hu Y M 2023 arXiv:2304.10340
- [12] Fan H-M, Hu Y-M, Barausse E, Sesana A, Zhang J-D, Zhang X, Zi T-G and Mei J 2020 *Phys. Rev. D* **102** 063016
- [13] Zhang X-T, Messenger C, Korsakova N, Chan M L, Hu Y-M and Zhang J 2022 *Phys. Rev. D* **105** 123027
- [14] Liang Z-C, Hu Y-M, Jiang Y, Cheng J, Zhang J-D and Mei J 2022 *Phys. Rev. D* **105** 022001
- [15] Cheng J, Li E-K, Hu Y-M, Liang Z-C, Zhang J-D and Mei J 2022 *Phys. Rev. D* **106** 124027
- [16] Seoane P A *et al* Laser interferometer space antenna 2023 *Living Rev. Rel.* **26** 2
- [17] Arun K G *et al* Laser interferometer space antenna 2022 *Living Rev. Rel.* **25** 4
- [18] Auclair P *et al* (The LISA Cosmology Working Group) 2023 *Living Rev. Rel.* **26** 2
- [19] Baker T *et al* (The LISA Cosmology Working Group) 2022 *J. Cosmol. Astropart. Phys.* JCAP08(2022)031
- [20] Bartolo N *et al* (The LISA Cosmology Working Group) 2022 *J. Cosmol. Astropart. Phys.* JCAP11(2022)009
- [21] Belgacem E *et al* (The LISA Cosmology Working Group) 2019 *J. Cosmol. Astropart. Phys.* JCAP07(2019)024
- [22] Babak S *et al* (Mock LISA Data Challenge Task Force) 2010 *Class. Quantum Grav.* **27** 084009
- [23] Baghi Q (LDC Working Group) 2022 The LISA data challenges 56th rencontres de moriond on gravitation (arXiv:2204.12142)
- [24] Ren Z, Zhao T, Cao Z, Guo Z-K, Han W-B, Jin H-B and Wu Y-L 2023 *Front. Phys.* **18** 64302
- [25] Arnaud K A *et al* 2007 *Class. Quantum Grav.* **24** S529
- [26] Babak S *et al* (Mock LISA Data Challenge Task Force) 2008 *Class. Quantum Grav.* **25** 114037
- [27] Gong Y, Luo J and Wang B 2021 *Nat. Astron.* **5** 881–9
- [28] Crowder J and Cornish N J 2005 *Phys. Rev. D* **72** 083005
- [29] Ruan W-H, Liu C, Guo Z-K, Wu Y-L and Cai R-G 2020 *Nat. Astron.* **4** 108–9
- [30] Wang G, Ni W-T, Han W-B, Yang S-C and Zhong X-Y 2020 *Phys. Rev. D* **102** 024089
- [31] Zhu L-G, Xie L-H, Hu Y-M, Liu S, Li E-K, Napolitano N R, Tang B-T, Zhang J-D and Mei J 2022 *Sci. China Phys. Mech. Astron.* **65** 259811
- [32] Lyu X, Li E K and Hu Y M 2023 arXiv:2307.12244
- [33] Liang Z-C, Li Z-Y, Cheng J, Li E-K, Zhang J-D and Hu Y-M 2023 *Phys. Rev. D* **107** 083033
- [34] Liang Z C, Li Z Y, Li E K, Zhang J D and Hu Y M 2023 arXiv:2307.01541
- [35] Schutz B F 2011 *Class. Quantum Grav.* **28** 125023
- [36] Torres-Orjuela A, Huang S J, Liang Z C, Liu S, Wang H T, Ye C Q, Hu Y M and Mei J 2023 arXiv:2307.16628
- [37] Xie Y, Chatterjee D, Holder G, Holz D E, Perkins S, Yagi K and Yunes N 2023 *Phys. Rev. D* **107** 043010
- [38] Roulet J, Olsen S, Mushkin J, Islam T, Venumadhav T, Zackay B and Zaldarriaga M 2022 *Phys. Rev. D* **106** 123015
- [39] Cutler C 1998 *Phys. Rev. D* **57** 7089–102
- [40] Marsat S and Baker J G 2018 arXiv:1806.10734 [gr-qc]
- [41] Marsat S, Baker J G and Dal Canton T 2021 *Phys. Rev. D* **103** 083011
- [42] Chen H Y, Lyu X Y, Li E K and Hu Y M 2023 arXiv:2309.06910
- [43] Hu X-C *et al* 2018 *Class. Quantum Grav.* **35** 095008
- [44] Rubbo L J, Cornish N J and Poujade O 2004 *Phys. Rev. D* **69** 082003 (arXiv:gr-qc/0311069)
- [45] Jolien D E and Creighton W G A 2011 *Gravitational waves Gravitational-Wave Physics and Astronomy* (Wiley) pp 49–95
- [46] Kogut A *et al* 1993 *Astrophys. J.* **419** 1
- [47] Goldberg J N, MacFarlane A J, Newman E T, Rohrlich F and Sudarshan E C G 1967 *J. Math. Phys.* **8** 2155

- [48] Yunes N, Arun K G, Berti E and Will C M 2009 *Phys. Rev. D* **80** 084001
- [49] Wahlquist H 1987 *Gen. Rel. Grav.* **19** 1101–13
- [50] Armstrong J W, Estabrook F B and Tinto M 1999 *Astrophys. J.* **527** 814–26
- [51] Katz M L, Bayle J-B, Chua A J K and Vallisneri M 2022 *Phys. Rev. D* **106** 103001
- [52] Estabrook F B and Wahlquist H D 1975 *Gen. Relativ. Gravit.* **6** 439–47
- [53] Romano J D and Cornish N J 2017 *Living Rev. Rel.* **20** 2
- [54] Cornish N J and Rubbo L J 2003 *Phys. Rev. D* **67** 022001
Cornish N J and Rubbo L J 2003 *Phys. Rev. D* **67** 029905 (Erratum)
- [55] Seto N 2020 *Phys. Rev. D* **102** 123547
- [56] Tinto M and Dhurandhar S V 2021 *Living Rev. Rel.* **24** 1
- [57] Prince T A, Tinto M, Larson S L and Armstrong J W 2002 *Phys. Rev. D* **66** 122002
- [58] Babak S, Petiteau A and Hewitson M 2021 arXiv:2108.01167
- [59] Nelemans G, Yungelson L R and Portegies Zwart S F 2001 *Astron. Astrophys.* **375** 890–8
- [60] Khan S, Husa S, Hannam M, Ohme F, Pürrer M, Jiménez Forteza X and Bohé A 2016 *Phys. Rev. D* **93** 044007
- [61] Husa S, Khan S, Hannam M, Pürrer M, Ohme F, Jiménez Forteza X and Bohé A 2016 *Phys. Rev. D* **93** 044006
- [62] Loutrel N 2020 arXiv:2009.11332
- [63] Huerta E A, Kumar P, McWilliams S T, O’Shaughnessy R and Yunes N 2014 *Phys. Rev. D* **90** 084016
- [64] LIGO Scientific Collaboration 2018 *Ligo Algorithm Library - Lalsuite Free Software (GPL)*
- [65] Ye C Q, Fan H M, Torres-Orjuela A, Zhang J d and Hu Y M 2023 arXiv:2310.03520
- [66] Katz M L, Chua A J,K, Speri L, Warburton N and Hughes S A 2021 *Phys. Rev. D* **104** 064047
- [67] Chua A J,K, Katz M L, Warburton N and Hughes S A 2021 *Phys. Rev. Lett.* **126** 051102
- [68] Chua A J,K, Galley C R and Vallisneri M 2019 *Phys. Rev. Lett.* **122** 211101
- [69] Allen B and Romano J D 1999 *Phys. Rev. D* **59** 102001
- [70] Rieck S, Criswell A W, Korol V, Keim M A, Bloom M and Mandic V 2023 arXiv:2308.12437
- [71] Allen B and Ottewill A C 1997 *Phys. Rev. D* **56** 545–63